Refinancing Inequality and Monetary Transmission Channel

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March 2026

Abstract

Mortgage refinancing is highly unequal across income groups and this inequality substantially weakens the transmission of monetary policy. I show that refinancing inequality appears in both the intensity and timing of refinancing: the bottom income quintile refinances at roughly two-thirds the rate of the top quintile and faces markedly longer delays, even when refinancing is financially beneficial. These gaps persist even after controlling for measures of lender credit tightness. The associated unrealized savings are large, exceeding 7.6% of monthly income for the bottom quintile. I develop a structural mortgage refinancing model that decomposes refinancing frictions into two distinct channels—inattention and hassle cost—and allows both frictions to vary by income state. The model matches the observed refinancing inequality, unlike homogeneous-friction benchmarks. Counterfactual experiments show that reducing frictions for lower-income households would amplify the effects of a policy rate cut by 13.7% in consumption over five years and reduce wealth inequality.

[*] I am deeply grateful to James Cloyne, Òscar Jordà, Nico Caramp, and Sanjay Singh for valuable comments and guidance. I used an AI language model to help improve grammar and phrasing in the manuscript. All substantive analyses, model specification, and empirical results were produced by the authors.
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1 Introduction

Mortgages play a key role in household finance due to their scale and prevalence. One of the most prominent features of the U.S. mortgage market is that it is dominated by long-term fixed-rate mortgages — about 96% of all outstanding mortgages are fixed-rate as of 2025[1]. This market feature emphasizes the role of refinancing as a transmission channel of monetary policy. For example, when the Federal Reserve lowers interest rates, mortgagors can benefit from refinancing into lower-rates, thereby freeing up cash flow to increase their consumption. At the same time, studies have identified significant frictions in refinancing (Keys et al. 2016; Andersen et al. 2020; Gerardi et al. 2023; Zhang 2022) and it could significantly impair the effectiveness of monetary policy (Beraja et al. 2019; Defusco et al. 2019; Defusco and Mondragon 2020).

Against this backdrop, I argue that heterogeneity in refinancing behavior is central for monetary policy transmission — and that income is a particularly important and underexplored dimension of this heterogeneity. Even holding fixed the average level of refinancing frictions, *who* refinances can be as important as *whether* refinancing occurs at all. If refinancing is systematically less likely among households with tighter liquidity constraints, then rate cuts deliver cash-flow relief precisely away from those with the highest marginal propensity to consume, weakening the real effects of monetary easing. Income is a particularly natural dimension along which to study this heterogeneity. It is a salient distributional margin, closely linked to liquidity constraints and MPC, yet income-group differences in refinancing behavior — and their aggregate transmission implications — remain relatively understudied. Existing work on heterogeneity in the refinancing channel has emphasized other margins — regional variation, housing equity, age, and contract features — while evidence on income-based differences has been largely episode-specific and has not been connected to aggregate consumption responses.

[1] Federal Housing Finance Agency (FHFA)

To motivate the empirical patterns, Figure 1 plots mortgage rates alongside prepayment activity by income quintile. Prepayment rises sharply when interest rates fall, consistent with refinancing-driven activity, but the response differs markedly across income groups despite a common aggregate interest-rate environment. Higher-income borrowers exhibit a much stronger prepayment response than lower-income borrowers.[2] Following **?**, I refer to this systematic income gradient in refinancing behavior as *refinancing inequality* — a pattern I examine at scale and connect explicitly to aggregate monetary policy transmission. The central question of the paper is whether — and by how much — this inequality dampens monetary policy transmission, and which refinancing frictions account for it.

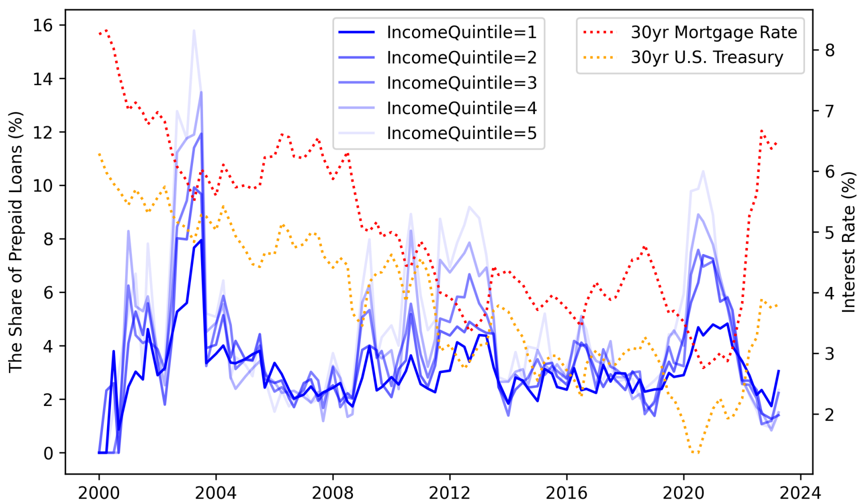


Figure 1: Prepayment Activity by Income Quintile

This paper proceeds in three steps. First, using loan-level panel data, I document substantial refinancing inequality across income groups, not only in the likelihood of refinancing but also in the timing of refinancing once households become in-the-money. Second, I show that this inequality extends to the transmission of monetary policy, with low-income households responding significantly less to interest rate cuts. Finally, I develop

[2]Income quintiles follow the same definition as in the main analysis: borrowers are assigned based on nationwide annual mortgage income distributions at origination, constructed from the Freddie Mac data.

a structural mortgage refinancing model with heterogeneous attention and execution frictions to interpret these patterns and quantify their macroeconomic implications.

I begin by documenting refinancing inequality using panel data from 2000 to 2022. The analysis reveals substantial refinancing inequality in both the *intensity* and *timing* of refinancing. When households are in-the-money, those in the lowest income quintile exhibit monthly refinancing probabilities that are roughly two-thirds of those in the top quintile. Decomposing refinancing behavior by the gap between the contracted mortgage rate and the market rate, I find that high-income households consistently display significantly higher refinancing probabilities than low-income households. Importantly, the inequality is most pronounced in the in-the-money region, indicating that even when refinancing is advantageous, low-income borrowers are far less likely to act. Beyond differences in refinancing probability, I provide novel evidence that refinancing inequality also appears in its *delays*. Measuring the share of loans that refinance after becoming in-the-money, I show that higher-income households refinance much more promptly and therefore capture financial gains earlier than their lower-income counterparts.

A natural concern is that these income-quintile gaps might be driven by lender-side underwriting. I therefore control for year-by-income mortgage rejection rates across all U.S. markets and a quarterly survey-based index of lending standards, and the refinancing gaps remain large and statistically significant. Moreover, robustness checks that stratify borrowers into credit-score quantiles show that, within any given credit-score group, refinancing inequality across income groups persists.

I further address concerns about income measurement and income mobility. Since income quintiles are assigned at origination, borrowers may have transitioned across quintiles by the time of refinancing. To assess the severity of this issue, I use the Panel Study of Income Dynamics (PSID) to quantify income-quintile transitions over a two-year horizon—approximately the average time to refinancing in my sample—and find that large cross-quintile movements are rare. To further validate the baseline results, I redefine income groups using the income quintile recorded on the new loan at the time of refinancing, restrict the sample to borrowers whose income quintile did not change between origination and refinancing, and subsample by loan age to limit the scope for income re-ranking. Across all of these exercises, the pattern of refinancing inequality remains intact. Additionally, I verify that the results are not driven by a particular time period by repeating the estimates across rolling five-year windows and by excluding the COVID-19 period, finding consistent evidence of inequality throughout.

To quantify how much lower-income households miss out on potential gains from refinancing, I estimate the economic magnitude of forgone savings, which are particularly large for low-income borrowers. By matching pre-refinance (old) and post-refinance (new) loans, I estimate the reduction in monthly mortgage payments due to refinancing. Households in the bottom and second-lowest quintiles could have saved approximately 7% of their monthly income had they refinanced when it was optimal to do so. This amount is substantial not only as a share of income but also because it translates into a persistent increase in monthly disposable income.

Beyond the empirical evidence, I develop a structural mortgage refinancing model to interpret the observed refinancing inequality and to quantify its macroeconomic implications. The key innovation of the model is to decompose refinancing frictions into two conceptually distinct channels—an inattention channel and an execution (hassle-cost) channel—and to allow both frictions to vary systematically across income states. This structure makes it possible to separately quantify the contribution of each channel to refinancing behavior. Attention heterogeneity is disciplined using a parsimonious parametric specification consistent with empirical patterns, while execution frictions are estimated non-parametrically across income groups. By allowing refinancing frictions to differ by income and by channel, the model successfully reproduces the observed refinancing inequality. In contrast, benchmark models in the existing literature that impose homogeneous refinancing frictions substantially overpredict refinancing among low-income households and underpredict it among high-income households.

The model further demonstrates that these frictions have meaningful implications for monetary policy transmission. A counterfactual experiment—raising the attention capacity for low-income households to the level observed among high-income borrowers—shows that a one-percentage-point interest rate cut would generate an additional 13.7% increase in aggregate consumption over five years, depending on the past interest rate paths. This amplification arises because higher attention enables more borrowers to refinance promptly when rates fall, thereby strengthening the pass-through of monetary policy to household cash flows and consumption. Since attention-related frictions are potentially mitigable through targeted policy interventions, the results highlight a realistic avenue through which monetary easing can be made more effective.

Related Literature This paper contributes to three strands of literature. The first strand is the literature on refinancing frictions and monetary policy transmission. A large body of work documents that households systematically under-refinance, attenuating the cash-flow channel of monetary policy (Keys et al. 2016; Agarwal et al. 2016; Di Maggio et al. 2017). Berger et al. (2021) and Eichenbaum et al. (2022) further show that the efficacy of the refinancing channel depends on the path of past interest rates, generating state-dependence in monetary transmission. Byrne et al. (2023) provide causal evidence from a field experiment that inattention constitutes the "last mile" blocking the pass-through of rate cuts to household consumption. Relative to this strand, I make two contributions. First, I show that not only the average level of refinancing frictions but also their *distribution* across income groups matters for monetary transmission: because the cash-flow benefits of rate cuts systematically bypass lower-income households with higher marginal propensities to consume, the aggregate consumption response is dampened in a way that average-friction models miss. Second, I decompose refinancing frictions into inattention and hassle cost components and identify which drives the income gradient in refinancing behavior. Consistent with Berger et al. (2021), I find that the attention channel dominates. This decomposition carries a direct policy implication: targeted interventions that raise attention among lower-income borrowers—of the kind studied experimentally by Byrne et al. (2023)—are likely to be particularly effective in reducing refinancing inequality and strengthening monetary pass-through.

The second strand examines heterogeneity in refinancing behavior. One line of work documents the sources and consequences of unequal refinancing, focusing on the phenomenon itself (Andersen et al. 2020; Gerardi et al. 2023; Agarwal et al. 2024) or on its redistributive implications (Berger et al. 2024; Fisher et al. 2024; Zhang 2022). A related line focuses on how institutional constraints generate refinancing failures that attenuate monetary transmission: Beraja et al. (2019) show that the regional distribution of housing equity dampens aggregate stimulus, while Di Maggio et al. (2020) document that GSE purchase eligibility shapes borrowers' ability to access lower rates. More broadly, studies that directly link *group-based* refinancing heterogeneity to the quantitative dampening of monetary transmission remain scarce, and the income dimension in particular—despite its direct connection to MPC heterogeneity and inequality—has been underexplored. Concurrent work by Jørring and Westphal (2025) uses survey evidence to document that refinancing take-up is systematically lower among high-MPC households, providing reduced-form evidence that the distribution of refinancing benefits matters for aggregate consumption. My paper complements their findings by embedding income-heterogeneous refinancing into a structural model, which allows me to go beyond documenting the pattern: I quantify the aggregate dampening effect on monetary transmission and conduct counterfactual policy experiments that are unavailable to survey-based approaches. Specifically, I develop a structural model with income-heterogeneous frictions—a dimension absent from Berger et al. (2021) and the broader literature—and show that a homogeneous-friction benchmark fails to replicate the observed inequality, establishing that the *distribution* of frictions across income groups, and not merely their average level, is necessary to match the data. This model then serves as the basis for counterfactual experiments that quantify, for the first time, how income-side friction heterogeneity dampens monetary transmission. I also provide the empirical foundation for this analysis by generalizing the income-heterogeneity findings of Agarwal et al. (2024): I document refinancing inequality not only in *intensity* but also in its *timing*—lower-income households refinance with significantly longer delays even after becoming in-the-money—extend the analysis from a COVID-era episode to a long panel covering more than twenty years, and show that the inequality survives controlling for lender-side credit tightness, and home equity, establishing income as an independent borrower-side friction dimension.

The third strand is the literature on monetary policy and inequality (Coibion et al. 2017; Kaplan et al. 2018; Auclert 2019; Amberg et al. 2022; Andersen et al. 2023). This paper contributes to this strand by providing a refinancing-based mechanism that directly links accommodative monetary policy to income inequality through unequal cash-flow pass-through: lower-income households, facing higher effective frictions, capture a systematically smaller share of the payment relief generated by rate cuts, widening the income gap in disposable income and, over time, in wealth accumulation. This mechanism also carries a policy implication: reforms that reduce refinancing frictions for lower-income borrowers—such as streamlining the application process or strengthening targeted outreach—could partially mitigate not only refinancing inequality itself but also its longer-run consequences for wealth inequality.

This paper briefly reviews the data and methodology in Section 2, then examines the average pattern of refinancing inequality over the entire period in Section 3 and provides a back-of-the-envelope analysis of potential savings that could have been gained by refinancing. In Section Appendix A, the focus shifts to analyzing the refinancing dynamics in response to interest rate changes. Finally, Section 4 develops a mortgage refinancing model with frictions to rationalize the empirical findings. The model successfully replicates the observed refinancing inequality and quantifies how relaxing these frictions amplifies the consumption response to interest rate cuts.

2 Institutional Background and Data

2.1 Effect of Income on Refinancing

The potential mechanisms by which income can influence refinancing can be outlined in three main ways. First, there is the friction caused by low liquidity and the presence of closing costs for low-income households. Closing costs typically include fees for the loan application, appraisal, title insurance, and other administrative services. It usually amounts to 2-5% of the loan balance, which can be a significant barrier to refinancing for households with low liquidity. If the borrower does not have immediate liquidity, it is usually possible to roll the closing costs into a new loan, though this presents a trade-off with the interest rate. The tradeoff entails deciding whether to pay higher upfront closing costs for a lower interest rate, reducing monthly payment over the loan term, or to opt for lower closing costs with a higher interest rate, resulting in higher monthly payments and total interest paid. Appendix Figure B.3 shows an example of the trade-off between upfront cost and interest rate. In this example, it can be observed that for the same property and loan, both cash-out (equity extraction) and non-cash-out refinancing options are associated with a lower interest rate when higher upfront costs are incurred.

Secondly, there is the Payment to Income (PTI) ratio regulation. In the United States, the PTI level is applied as a soft borrowing constraint but is incorporated into the loan approval process by setting a standard. For example, the standard PTI level for a conventional loan is 36%, and a higher PTI level can only be approved if exceptional conditions such as a low LTV and a high credit score are met. As can be seen in actual data in Table B.7, low-income households exhibit higher PTI at loan origination compared to high-income households.

Therefore, low-income households are more likely to hit the PTI limit than high-income households given the same distribution of idiosyncratic income shocks.

Lastly, there is the association with inattention. As shown by Andersen et al., inattention, which is a significant cause of observable suboptimal refinancing, is positively correlated with income. Consequently, through behavioral issues, different income groups may exhibit varying refinancing tendencies.

2.2 Data Overview

The primary dataset used in this study is the Freddie Mac Single Family Loan-Level Dataset, which covers loans originating from 2000 to 2022. This dataset provides comprehensive information on both loan issuance and performance.[3] The issuance data includes a rich set of loan and borrower characteristics at the time of origination. Key variables include the initial loan balance, loan term, interest rate, credit score, Loan-to-Value (LTV) ratio, Payment-to-Income (PTI) ratio, ZIP code, and estimated income. These variables allow for detailed analysis while controlling for factors influencing refinancing decisions, and I restricted sample to fixed-rate mortgages. In addition to issuance data, the performance data offers monthly observations of loans, which include the remaining balance, current interest rate, amortization, and prepayment status. This panel data enables the tracking of loan performance over time, facilitating a dynamic analysis of refinancing behavior.

The income for each loan can be back-calculated using Freddie Mac's issuance data, using the original interest rate, loan amount, maturity, and PTI information. Based on the distribution of this derived income of each origination year, the loans were then divided into quintiles.

[3] The Freddie Mac data set represents approximately 20%-30% of the total U.S. mortgage market, with the notable exclusion of "jumbo" loans, which are larger than the conforming loan limits set by the Federal Housing Finance Agency every year.

To complement the Freddie Mac data, this study also utilizes data from the Home Mortgage Disclosure Act (HMDA) from 2000 to 2022.[4] The HMDA data consists of detailed records on individual mortgage applications and originations collected by financial institutions. The data include a wide array of information, such as the loan amount, type of loan, property location, borrower characteristics, and the outcome of the application. I matched Freddie Mac to HMDA using common variables to find corresponding loans, and used information from HMDA for analysis.

Detailed summary statistics of the final panel used in the analysis can be found in Appendix subsection B.1.

2.3 Identifying Refinancing Loans

The main challenge when attempting to identify refinancing activity from the Freddie Mac loan panel is it only reports prepayment activity without a reason. As prepayment of mortgages could happen due to various other reasons than refinancing, such as moving or trading of houses, identifying refinancing is a key step for this analysis. I match a prepaid loan to a new loan also funded by Freddie Mac originated within a 45-day window of the closure of the prepaid loan, following **?**.[5]

However, during the matching process, the restricted regional data of the publicly available Freddie Mac panel posed a significant challenge. The Freddie Mac data only provides three-digit ZIP codes for properties, which cover an excessively large area, leading to numerous duplicate matches and making unique matching infeasible. To address this issue, I matched the Freddie Mac data with HMDA data, utilizing the finer regional

[4]The Home Mortgage Disclosure Act (HMDA) requires financial institutions to maintain and annually disclose data about home purchases, home purchase pre-approvals, home improvement, and refinance applications. The HMDA data captures the majority of mortgages in the U.S., accounting for about 92 percent of originations nationally in 2017, according to the Consumer Financial Protection Bureau.

[5] The 45-day window is based on the institutional background of the rate lock period. A rate lock is an agreement between the borrower and the lender that guarantees a specified interest rate for a predetermined period while the mortgage application is being processed. The 45-day period is a common duration within this range. Thus, when a loan is prepaid for refinancing, a new loan is highly likely to be originated within this window.

index at the census tract level provided by HMDA. Given the potential for bias in the matching processes, I performed a series of data validation checks, as displayed in the Appendix subsection B.3, and found no significant errors in the matched sample compared to the population.

2.4 Identifying In-the-money Loans

The decision for households to refinance is determined by the two types of incentives: first, the cash-out motive due to liquidity needs in response to idiosyncratic shocks, and second, the demand for reducing financing costs through lower mortgage rates. The first refinancing incentive is known to be insensitive to interest rates, while the second incentive is directly affected by monetary policy.[6] Since the focus of this study is to analyze the differences in the transmission of the benefits of accommodative monetary policy across income groups, the second incentive, the motive for interest rate reduction, should be emphasized. Therefore, distinguishing whether there is a financial benefit from lowering mortgage rates during refinancing is one of the key identifications. An in-the-money option refers to an option that provides a financial benefit when exercised. Therefore, a fixed-rate mortgage becomes an in-the-money loan when refinancing is financially advantageous.

$$\text{In-the-money } if \;\; m_i^* - \tilde{m}_{it} > \theta_{it} \tag{1}$$

The financial benefit of refinancing arises when the borrower's current mortgage rate exceeds the rate available on a newly originated loan, that is, when the loan is "in-the-money." Formally, a loan is identified as in-the-money when the contract rate m_i^* is higher than the prevailing market rate m_{it} by more than the threshold θ_{it}, as shown in Equation 1. The key object in this condition is the rate incentive: the gap between the

[6] Chen et al. (2020) focused on the cash-out motive of refinancing and suggested that refinancing decisions of low-liquidity households are insensitive to interest rate changes.

borrower's current coupon and the rate they could obtain today. To construct a clean measure of this incentive, I estimate the obtainable rate m_{it} as the median origination rate among newly originated loans within a granular borrower–loan characteristic cell defined by origination quarter, ZIP code, credit score bucket, loan-to-value ratio, and income quintile, following Defusco and Mondragon (2020). This cell-based approach compares each borrower to observationally similar borrowers who actually refinanced in the same local market at the same point in time, yielding a benchmark rate that reflects what the borrower could plausibly obtain given their characteristics. Crucially, this measure of the rate incentive is held fixed when comparing refinancing behavior across income groups, so that any differences in take-up rates across the income distribution reflect heterogeneity in the response to a given incentive rather than differences in the incentive itself.

In the baseline specification, I set $\theta_{it} = 0$, assuming that refinancing is financially beneficial whenever the contracted rate exceeds the estimated market rate. As a robustness check, I also employ a more conservative identification rule based on the Agarwal-Driscoll-Laibson gap (hereafter *ADL gap*), where $\theta_{i,t} = \theta_{it}^{ADL}$ corresponds to the optimal refinancing threshold derived from Agarwal et al. (2013). This threshold incorporates refinancing costs, expected exogenous prepayment, remaining maturity, and interest rate volatility, thereby tightening the condition under which a loan qualifies as in-the-money. Although more restrictive, this ADL-based rule yields qualitatively similar patterns of refinancing inequality. The detailed formula of the ADL gap is described in Appendix D.

3 Evidence of Refinancing Inequality

3.1 Average Refinancing Probability

I begin by identifying varying refinancing probability across income groups using a linear probability model (LPM).[7] In this regression estimation, I analyze only the loan observations that were in-the-money. The reason for this is that the focus of this study is on whether interest rate reductions lead to heterogeneous benefits across income groups, so refinancing unrelated to interest rate reductions is not of interest.[8] Specifically, I used the following specification:

$$Refi\_Indicator_{it} = \alpha + \sum_{q=2}^{5} \beta_q \cdot IncomeQuintile_{iq} + \Gamma' \cdot X_{it} + \nu_{z(i)} + \lambda_{v(i)} + \epsilon_{it} \quad (2)$$

$Refi\_Indicator$ is a binary variable that equals 1 if the loan is refinanced and 0 otherwise. j denotes income quintiles 2 through 5. X_{it} represents a set of control variables, and $\nu_{z(i)}$ and $\lambda_{v(i)}$ are ZIP code and vintage year-quarter fixed effects, respectively. Therefore, the β_j means the refinancing propensity of income quintile j compared to the bottom quintile, conditional on being in-the-money.

One concern is that the difference in refinancing probabilities across quintiles could be driven by other factors. To control for loan and borrower characteristics that are influential in refinancing, I used a rich set of control variables of credit score, loan age, loan balance to income ratio, ZIP code, and original vintage year, following literature. Also, the control variables were included as categorical variables to allow for non-parametric regression.[9]

However, existing research has not controlled for two other important variables that affect refinancing by income level. First, as highlighted in the literature by Beraja et al.

[7] The LPM assumes a linear relationship between the dependent variable and the explanatory variables, but since I allowed for nonlinearity by using category variables for income quintiles, I found no significant differences compared to the Logit results.

[8] For example, households who face idiosyncratic income shocks would refinance to extract equity regardless of interest rate, such as Chen et al. (2020) documented.

[9] I used categorical variables for LTV, FICO, loan age, loan balance, and estimated LTV.

(2019), the equity distribution channel could drive heterogeneous refinancing activities by income group. For example, a low-income borrower would be more likely to purchase a home in a low-income area. If low-income areas are more susceptible to a decline in home prices during an economic downturn, the likelihood that the low-income borrower will become underwater increases, resulting in a lower ability to refinance. Such dynamic changes cannot be controlled by regional fixed effects. To address this, I controlled for the equity channel by estimating the loan-to-value (LTV) ratio for all loans on a monthly basis. Starting with the LTV and loan size at origination, I derived the collateral value and tracked home values monthly using the Zillow ZIP code-level house price index. By combining the estimated home value with the observed remaining loan balance from the loan performance data, I was able to estimate the LTV for each loan each month. I controlled for the Home Affordable Refinance Program (HARP) in the same context. HARP was implemented from 2009 to 2018 to help borrowers with little or no equity in their homes refinance into more affordable mortgages. Because I matched the pre-refi and post-refi loans, I was able to observe whether the new loan was refinanced through HARP, allowing me to control for the policy factor that affects the equity distribution channel.

Another potentially important factor that could lead to refinancing inequality is the differential impact of lender-side credit supply tightness across income groups. Just as credit supply expanded disproportionately for low-income households during periods of credit expansion (Mian and Sufi, 2009), tightening credit supply could differentially affect low-income borrowers. To address this issue, I include two control variables. First, quarterly changes in the credit tightness index from the Federal Reserve's Senior Loan Officer Survey were used. Second, I controlled for loan application denial rates by income quintile derived from Home Mortgage Disclosure Act (HMDA) data. Using the same income quintile boundaries that were used in the main loan panel, HMDA applications were divided into five income groups, and I used annual changes in loan application denial rates of each income quintile.

Table 1: Average Refinancing Rate by Income Quintile (Simple Rate Gap)

Dependent	Refinancing Indicator × 100			
Equation	(1)	(2)	(3)	(4)
IncomeQuintile 2	0.213[a]	0.316[a]	0.324[a]	0.296[a]
	(0.133)	(0.115)	(0.117)	(0.131)
IncomeQuintile 3	0.072[a]	0.070[a]	0.077[a]	0.080[a]
	(0.115)	(0.117)	(0.117)	(0.131)
IncomeQuintile 4	0.809[a]	0.647[a]	0.649[a]	0.470[a]
	(0.115)	(0.117)	(0.117)	(0.131)
IncomeQuintile 5	1.452[a]	1.573[a]	1.672[a]	1.153[a]
	(0.133)	(0.115)	(0.117)	(0.131)
Constant	1.923[a]	0.421[a]	0.504[a]	0.562[a]
	(0.064)	(0.199)	(0.218)	(0.215)
Control variables				
LTV, credit, loan age, rate inc., ZIP		x	x	x
Credit tightness, HARP			x	x
Loan balance, Est. LTV				x
Observations	1,342,423	1,342,423	1,342,423	1,342,423
R-squared	0.002	0.010	0.011	0.011

Notes: Superscript a, b, c denote significance at 1%, 5%, 10% levels respectively. S.E. in parenthesis. Standard errors are clustered by zipcode and original vintage year.

Table 2: Average Refinancing Rate by Income Quintile (ADL gap)

Dependent	Refinancing Indicator × 100			
Equation	(1)	(2)	(3)	(4)
Income Quintile 2	0.518[a]	0.562[a]	0.577[a]	0.579[a]
	(0.065)	(0.068)	(0.075)	(0.072)
Income Quintile 3	0.924[a]	1.030[a]	1.064[a]	0.923[a]
	(0.093)	(0.107)	(0.113)	(0.100)
Income Quintile 4	1.321[a]	1.472[a]	1.501[a]	1.148[a]
	(0.120)	(0.125)	(0.131)	(0.118)
Income Quintile 5	1.872[a]	1.909[a]	2.057[a]	1.571[a]
	(0.139)	(0.140)	(0.151)	(0.135)
Constant	1.940[a]	1.199[a]	1.317[a]	1.05[a]
	(0.064)	(0.199)	(0.218)	(0.215)
Control variables				
LTV, credit, loan age, rate inc., ZIP		x	x	x
Credit tightness, HARP			x	x
Loan balance, Est. LTV				x
Observations	1,342,278	1,342,278	1,342,278	1,342,278
R-squared	0.001	0.013	0.014	0.014

Notes: Superscript a, b, c denote significance at 1%, 5%, 10% levels respectively. S.E. in parenthesis. Standard errors are clustered by zipcode and original vintage year.

The baseline result of this regression using the simple rule of threshold ($\theta_{it} = 0$) is reported in Table 1. Column (a) shows the unconditional probability, indicating that the highest income quintile has a 1.45 percentage points higher monthly refinancing probability compared to the bottom quintile. As control variables are gradually added, the final base model in column (4) shows a 1.15 percentage points difference.

The predicted refinancing probability by income quintile, computed from the same regression estimates, is shown in Figure 2. The predicted probability illustrates the impact of differences in income quintiles on the final dependent variable when all control variables are at their mean values. Although there are some variations across quintiles with the addition of control variables, it is evident that the predicted probability increases clearly with income.

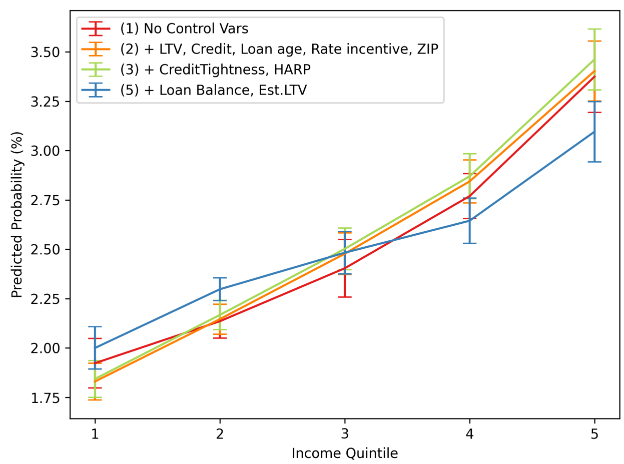


Figure 2: Predicted Refinancing Probability by Income Quintile

Notes: This plot shows the predicted monthly refinancing probability conditional on being in-the-money, using Equation 2. The 95% confidence intervals are represented by error bars and each line corresponds to the columns of the regression result in Table 1. Standard errors are clustered by ZIP code and original vintage year.

As shown in Table 2, the same regression is estimated using the *ADL gap*, which tightens the definition of being *in-the-money*. As previously discussed, this approach identifies refinancing profitability in a more conservative manner, requiring a larger gap between the coupon rate and the prevailing market rate compared with the simple threshold rule.

Consequently, the estimated refinancing probabilities are higher than those in Table 1. Even under this stricter identification, high-income households exhibit significantly higher refinancing propensities than low-income households, with all coefficients statistically significant at the 1% level. Overall, after controlling for all covariates, households in the top income quintile show monthly refinancing probabilities of around 1.7-2.5%, while the bottom-quintile households refinance at least 50% less frequently, with a gap of roughly 1-1.1 % points.

3.2 Refinancing Probability by Rate Gap

To further substantiate the average refinancing probabilities, I decompose the observations by interest rate gap and compare refinancing activity across income groups.

As discussed in subsection 2.4, I identify in-the-money in two ways: loans based on the rate gap between each loan's current coupon rate and the prevailing market rate on newly originated loans with comparable characteristics (ZIP code, credit score, LTV, and income). In the baseline specification, I apply a simple rule that sets the refinancing threshold to zero, effectively assuming no upfront costs. As a robustness check, I also adopt the Agarwal et al. (2013) optimal refinancing rule, which endogenizes the threshold by incorporating closing costs, loan maturity, expected prepayment behavior, and interest rate volatility. This ADL-based threshold provides a more realistic measure of refinancing gaps while preserving comparability with the baseline setup.

$$Refi\_Indicator_{it} = \alpha + \sum_{Quintile \in Q} \sum_{Bin \in K} \beta_q^k \cdot \mathbf{1}[RateGap_{it} \in k] \cdot \mathbf{1}[IncomeQuintile_i \in Q] \tag{3}$$

$$+ \Gamma' X_{it} + \nu_{z(i)} + \lambda_{v(i)} + \epsilon_{it} \tag{4}$$

I controlled the same variables as the previous regression using a non-parametric regression as specified in Equation 3. Again, $Refi\_Indicator_{it}$ is a indicator variable equal to 1 when the loan i is refinanced in month t. k denotes rate gap bins and j is the five income quintile at the time of origination. $\mathbf{1}[RateIncentive_{it} \in k]$ is a dummy variable equal to 1 when the rate gap falls in bin k, and X_{it} is a series of control variables. $\nu_{z(i)}$ and $\lambda_{v(i)}$ are ZIP code and vintage year-quarter fixed effects, respectively, and standard errors clustered by zip code and original vintage year.

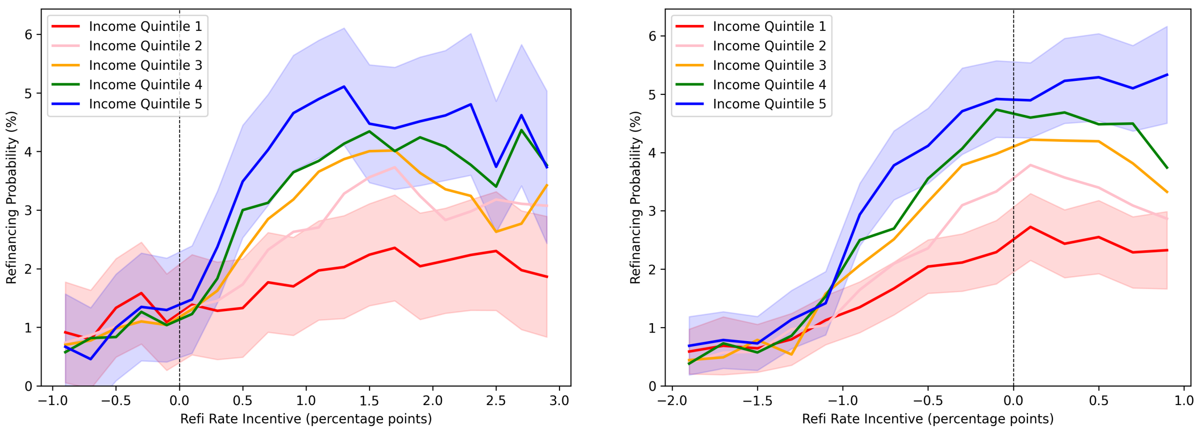
The left panel of Figure 3 presents the refinancing probability estimated using the simplest identification rule, where the threshold is set to zero. As shown, in the out-of-the-money region, refinancing rates are broadly similar across income quintiles. However, once loans enter the in-the-money region, a clear heterogeneity emerges. Even after accounting for the confidence bands of the first and fifth quintiles, high-income households are significantly more likely to refinance than low-income ones.

The right panel applies a more robust identification method based on the *ADL gap*, which allows the refinancing threshold to vary by loan and time (i, t). It is worth noting that this plot does not simply represent a horizontal shift of the left panel; rather, the bin assignment changes dynamically for each observation as the threshold evolves. This conservative specification already reveals refinancing inequality beginning around minus one percentage point, and the gap widens further in the in-the-money region.

Overall, regardless of the specific identification method, the pattern of refinancing inequality is consistently observed across nearly all rate-gap bins.

3.3 Delays in Refinancing Decisions

In the previous regression analysis, I focused on the *intensity* dimension of refinancing inequality—how much borrowers refinance when it is financially beneficial to do so. Most of the existing literature on heterogeneous refinancing behavior has centered on this margin. However, because refinancing is an option-like decision, when borrowers choose to



(a) Simple Rate Gap ($\theta_{it} = 0$) (b) ADL Gap ($\theta_{it} = ADLgap_{it}$)

Figure 3: Refinancing Inequality by Rate Gap

exercise the option is an equally important source of heterogeneity. The *timing* of refinancing may generate meaningful differences in realized savings even among borrowers with similar incentives.

To examine this timing margin, I align all loans at the month in which they first become in-the-money and compute, for each income quintile, the share of loans that remain outstanding over time. This approach yields a monthly survival rate of outstanding mortgages relative to the in-the-money point.

To implement this approach cleanly, I account for an important data feature. A complication arises because a loan may fall out-of-the-money shortly after first becoming in-the-money. In such cases, I treat these in-the-money occurrences as part of the same episode as long as the time gap between consecutive in-the-money months is less than twelve months. For example, suppose a loan becomes in-the-money and begins an episode, remains in that state for several months, then briefly falls out of the money, and becomes in-the-money again six months later. I treat this entire sequence as a single in-the-money episode, since the interval between the two in-the-money observations is less than twelve months.

In this analysis, I use a more robust measure of being in-the-money based on the ADL gap, as described in subsection 2.4. This measure is particularly important for analyzing refinancing timing, because the incentive to refinance declines sharply as a mortgage approaches maturity. The ADL-based measure incorporates this feature and therefore provides a more accurate basis for evaluating refinancing behavior over time.

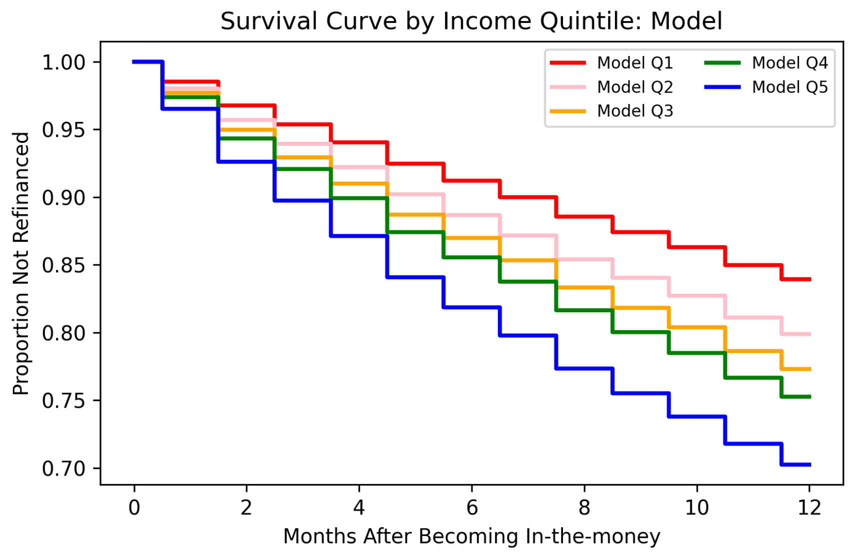


Figure 4: Proportion of Remaining Loans by Months After Becoming In-the-money

Figure 4 presents the resulting survival curves. Starting about two months after entering the money, loans held by households in the top income quintiles exhibit noticeably lower survival rates—meaning they refinance more quickly—than those in the bottom quintiles. This timing gap is persistent and economically meaningful. For example, the top quintile reaches a 50 percent survival rate after roughly seven months, whereas the bottom quintile takes about nine months. At the 40 percent threshold, the difference is even larger: ten months versus fifteen months. This analysis highlights a dimension of refinancing behavior that has received relatively little attention in the literature. To my knowledge, this is the first empirical evidence documenting a significant refinancing delay among lower-income households, complementing the existing focus on differences in refinancing intensity.

3.4 Robustness Checks

Concerns may remain that the observed refinancing inequality is driven primarily by lender-side underwriting—rather than borrower-side refinancing behavior—if low-income borrowers face systematically tighter approval constraints at the time of refinancing. While the baseline specification already controls for a rich set of borrower and loan characteristics, I further assess this channel by re-estimating the baseline regression within credit-score quartiles, a key underwriting input. The results show that the income gradient in refinancing remains pronounced within each credit-score bin: higher-income borrowers refinance at substantially higher rates than lower-income borrowers, even among borrowers with similar credit scores (see Figure C.7).

I also examine whether affordability constraints can explain the gradient by splitting the sample into payment-to-income (PTI) bins corresponding to common underwriting thresholds. Re-estimating the baseline specification within each PTI bin yields a similar pattern: the estimated income-quintile differences are broadly stable across PTI ranges, and remain statistically meaningful except in bins with limited observations. Importantly, the income gradient persists even in the most underwriting-favorable group with PTI below 36 percent, where approval constraints should be least binding (see Table C.10).

Another major concerns is that since income quintiles are observed at the time of origination, changes in income quintiles at the time of refinancing might cause bias in regression results. Given that income quintile is a key variable, I performed four different robustness checks to validate the previous results.

First, I calculated the average loan age up to the refinancing point and used the Panel Study of Income Dynamics (PSID) data to track changes in mortgagors' income quintiles over the same period. In my sample, the average time to refinancing was 23 months. Therefore, I examined the changes in mortgagors' income quintiles over two years using the same yearly-changing income quintile thresholds. The probability of the lowest income quintile rising to the highest quintile after two years was 5.6%, while the probability of the highest quintile falling to the lowest quintile was 2.4%, both very low as suggested in **??**[10]. Although the data's limitations prevent tracking income changes, these results suggest that my estimates are unlikely to be significantly distorted by frequent changes in income quintiles.

Second, my data panel matches old loans with new loans, allowing for the observation of income and other characteristics at the time of refinancing. Therefore, I was able to perform the same regression using Equation 2 with income quintile and other control variables at the time of refinancing. The results confirmed that the pattern consistent with the benchmark regression was maintained. As shown in **??**, even with an increase in control variables, the predicted refinancing probability for high-income households remained significantly higher than for lower income quintile groups[11].

Third, I conducted a regression analysis on only those loans where there was no change in income quintile between the new and old loans. The results still showed a clear refinancing inequality, as illustrated in Table C.12. This indicates that the previous results hold even when excluding changes in income quintile.

Lastly, since the probability of changes in a borrower's income quintile is lower for loans with shorter ages, I conducted repeated regression analyses by varying the age of the sample loans. The results showed a clear refinancing inequality regardless of the loan age cut. Table C.13 illustrates the predicted probabilities from regression (5), which includes all control variables, analyzed by progressively widening the loan age range: less than one, two, three, five, and seven years. As the loan age sample increased from one year or less to two years or less, there was an overall increase in probability, causing a line

[10] Broadening the scope, the probability of the bottom two (1st and 2nd) income quintiles remaining in the same two quintiles after two years was 72.7%, whereas the probability of changing to the top two (4th and 5th) quintiles was 12.6%. Conversely, the probability of the top two income quintiles remaining in the same quintiles was 81.6%, while the probability of falling to the bottom two quintiles was 8.2%, indicating a low likelihood.

[11] Note that this result is estimated based only on matched loans, excluding loans that were never refinanced. Therefore, the monthly refinancing probability is higher than in the benchmark regression. Also, the HARP variable, when based on new loans, showed that all loans were concentrated in quintiles 1 and 2. Controlling for this variable eliminated the differences between income quintiles. Therefore, it was not possible to control for the HARP variable.

shift. However, it was still evident that higher income was significantly associated with higher refinancing probabilities. The same pattern persisted when increasing the loan age limit further.

Other robustness checks using the Survey of Consumer Finance (SCF) can be found in the Appendix subsection C.3.

3.5 Unrealized Savings from Refinancing

Having identified significant refinancing inequality across income groups, I now investigate the extent to which this results in missed increases in disposable income. This study aims to examine how refinancing inequality ultimately impacts the monetary policy transmission channel, making the quantification of the saving amount crucial. I conducted the analysis on how much households' disposable income could have increased conditional on refinancing.

To start this analysis, I switched to a matched pair dataset of old and new loans. I analyzed how much each income group could reduce their interest rates through refinancing using a linear probability model. Using Equation 2, I changed the dependent variable as the difference between the interest rate of the new loan and that of the old loan, leading to the specification of Equation 5. I restrict the sample to in-the-money observations, since the calculation of missing gains is only meaningful when refinancing would have generated a financial benefit.

$$OldRate_i - NewRate_i = \alpha + \sum_{q=2}^{5} \beta_q \cdot IncomeQuintile_{ij} + \Gamma' \cdot X_{it} + \nu_{z(i)} + \lambda_{v(i)} + \epsilon_{it} \quad (5)$$

Based on the estimated rate reductions, I performed a back-of-the-envelope calculation to estimate the potential savings. By subtracting the rate reduction attributed solely to income quintiles from the existing interest rates, I derived the new post-refinancing interest rates. Assuming all refinancing are done with a 30-year fixed-rate mortgage, I calculated

Table 3: Income Gradient in Refinancing Within PTI Bins (ITM Sample)

Income Quintile	Avg. Income ($)	Avg. Balance ($)	Est. Rate Gap (pp)	Mthly. Pymt. Reduction ($)	Reduction / Income (%)
1	16,707	91,842	1.07	104.18	7.61
2	28,175	147,917	1.06	164.73	6.79
3	39,441	197,424	1.05	217.76	6.33
4	54,884	256,552	1.05	283.44	5.76
5	97,976	316,852	1.07	409.54	4.37

Notes: Payment savings assume refinancing resets the loan to a 30-year FRM term. Rate gaps are estimated using Eq. (5) (full controls). Freddie Mac statistics suggest that about 70–75% of refinances keep the term length unchanged. Sample is restricted to in-the-money observations ($gap > 0$).

the monthly payment reduction using these new estimated rates and examined this as a proportion of income.

The results Table 3 shows that the bottom two income quintiles could have saved 6.8-7.6% of their monthly income through refinancing. This is a significant amount, as it represents additional disposable income occurring every month, not just a one-time saving. Although the absolute dollar amount of missed gains can be larger for higher-income borrowers—primarily because they tend to hold larger mortgage balances—the foregone savings constitute a much larger share of income for lower-income mortgagors. Given their higher marginal propensity to consume, the missing gains for these households likely translate into substantial unrealized consumption responses.

As a robustness check, I also estimate unrealized savings using an alternative approach: for each in-the-money loan-month, I construct a counterfactual refinancing rate by subtracting the income-group-specific predicted rate gap—estimated from Equation 22—from the borrower's current coupon, and translate the implied payment reduction into dollars using loan-specific balance and maturity. This alternative procedure yields similar estimates, with the bottom quintile facing foregone monthly savings of approximately 7.6% of income, corroborating the findings from the matched-loan approach. Together, these findings suggest that refinancing inequality could materially weaken the pass-through of monetary policy to disposable income and consumption, precisely among households with the highest marginal propensity to consume. (see subsection C.4 for details).

4 A Model for Heterogeneous Refinancing Friction

To quantitatively assess how the refinancing inequality patterns documented in the empirical analysis affect the transmission of monetary policy, this paper develops a structural model of household mortgage refinancing. The model builds on the framework of Berger et al. (2021) , but departs from the baseline specification in an important dimension. In the original model, refinancing inequality does not emerge: instead, lower-income households—due to their limited liquid asset holdings—behave as wealthy hand-to-mouth agents and participate more actively in refinancing, generating a pattern opposite to that observed in the data. By incorporating three additional mechanisms described below, the proposed model successfully reproduces the empirically documented refinancing inequality, whereby higher-income households refinance more frequently while lower-income households exhibit systematically lower refinancing intensity.

The model improves upon the existing framework along three key dimensions. First, while the baseline model relies solely on an inattention channel to rationalize low refinancing rates, this paper distinguishes between two conceptually different sources of refinancing frictions: inattention and hassle costs. Separating these frictions allows the model to capture distinct behavioral and economic mechanisms underlying refinancing decisions. Second, the baseline specification models attention as a simple on–off switch depending on whether a household is in-the-money, which fails to capture how attention varies with the magnitude of the refinancing incentive. This limitation is quantitatively important, as the strength of monetary policy transmission may depend on the size of interest rate cuts. To address this, the model introduces a gap-dependent attention intensity rather than a binary attention mechanism. Third, and most importantly, the model allows both inattention and hassle cost frictions to vary by income state. Attention heterogeneity is disciplined using a parsimonious parametric structure consistent with empirical patterns observed in the data, while refinancing thresholds associated with hassle costs are estimated fully nonparametrically.

4.1 Model setup

The economy is populated by a continuum of risk-averse households who solve an infinite-horizon optimization problem in continuous time. Households face both aggregate interest rate risk and idiosyncratic labor income shocks. The aggregate short rate follows an exogenous process and is controlled by the monetary authority. Households are heterogeneous with respect to labor assets, outstanding mortgage balances, coupon rates, and income. Each household holds a long-term fixed-rate mortgage and can choose whether to refinance, including the option to extract home equity through cash-out refinancing. Refinancing requires paying a fixed cost and, in addition, households face two state-dependent refinancing frictions that are described below. On the supply side, lenders are risk-neutral. Taking the exogenous short rate as given, lenders price mortgage contracts based on expected prepayment behavior summarized by endogenous hazard rates, and supply mortgages such that equilibrium mortgage rates clear the mortgage market

4.2 Mortgage Refinancing and Frictions

Mortgage refinancing in the model is governed by two distinct and complementary frictions, which I explicitly *decompose* to isolate their separate economic roles. The first is a *threshold-type friction*, capturing the fact that refinancing entails not only explicit monetary fixed costs but also non-monetary costs, such as time costs, information-processing costs, and search costs for better mortgage terms. If this friction were the only impediment to refinancing, households would refinance with probability one whenever the gains from refinancing exceed a sufficiently high threshold.

The second friction is *inattention*. Even when refinancing gains are large enough to overcome threshold costs, observed refinancing rates remain far below one. This pattern suggests that households do not continuously monitor refinancing opportunities, but instead pay attention only intermittently. The model captures this behavior through a Calvo-style attention friction.

By decomposing refinancing frictions into an execution margin (threshold) and an attention margin (frequency), the model allows us to separately quantify the contribution of each channel to refinancing activity and to conduct counterfactual exercises that shut down individual frictions. This decomposition is central to understanding how refinancing inequality shapes the transmission of monetary policy.

Hassle-Cost Channel The threshold-type friction is interpreted as a hassle-cost (execution) channel, reflecting the costs associated with executing a refinancing transaction. Information-processing and search costs depend on financial sophistication and education, while the time required to evaluate and complete refinancing entails an opportunity cost that increases with income. As a result, the effective hurdle to refinancing may vary systematically across income groups.

Formally, the hassle-cost channel is modeled as an income-dependent refinancing threshold. A household executes a refinancing transaction only if the value gain from refinancing exceeds an income-specific threshold. To avoid imposing restrictive functional form assumptions, these thresholds are estimated fully nonparametrically across income groups. The resulting execution rule is summarized in equation (6), which characterizes refinancing as a comparison between the value gain from refinancing and the income-dependent hassle-cost threshold.

$$\rho^{refi} = \begin{cases} 1, & \text{if } V^{\text{refi}} - V^{\text{stay}} > \Phi_s \\ 0, & \text{otherwise.} \end{cases} \tag{6}$$

Attention Friction The attention channel governs how frequently households pay attention to refinancing opportunities over calendar time. Existing models typically assume that attention switches discretely on or off depending on whether a household is in-the-money. However, this binary formulation fails to capture a key behavioral feature of refinancing behavior: households naturally become more attentive to refinancing as the gap between their current mortgage coupon rate and the prevailing market rate widens. This distinction is quantitatively important, as the strength of monetary policy transmission may depend on the magnitude of interest rate movements rather than solely on whether refinancing is profitable.

In the model, attention varies continuously with the refinancing incentive, measured by the interest rate gap. Households receive opportunities to consider refinancing according to a Poisson process, whose arrival intensity depends on both the interest rate gap and the household's income state. Attention intensity increases as the household becomes deeper in-the-money.

The functional form of attention is disciplined by two robust empirical patterns observed in the data. First, even for households that are deep in-the-money, attention to refinancing remains limited rather than approaching one. Second, the maximum level of attention differs systematically across income groups. These features are captured by a parsimonious, gap-dependent attention specification, formalized in equation (7), which allows for income-dependent heterogeneity in maximum attention capacity.

$$\chi(gap_t, s_t) = \chi_{\max}(s_t) \cdot \frac{1}{1 + \exp\left(-k\,[\,gap_t - \mu\,]\right)} \tag{7}$$

Here, $gap \equiv m_t^* - m_t$ denotes the difference between the household's current mortgage coupon rate (m_t^*) and the prevailing market mortgage rate (m_t). $\chi_{\max}(s_t)$ denotes the maximum attention capacity for households in income state s, capturing the upper bound on how frequently a household can pay attention to refinancing opportunities even when it is deep in-the-money. The parameter μ governs the location of the attention response, representing a region in which refinancing incentives are too small to meaningfully trigger attention. The slope parameter k controls how sensitively attention responds to changes in the interest rate gap, that is, how rapidly attention intensity increases as the gap widens. Since the mortgage coupon rate is fixed, movements in the interest rate gap are driven by changes in the market interest rate, so k captures how households' refinancing attention responds to monetary policy–induced declines in market rates.

4.3 Household Problem

Households maximize lifetime utility by choosing both their consumption (C_t) path, the mortgage refinancing (ρ_t^{refi}), and cash-out decision (ΔF_t^{co}). These decisions are made subject to the structure of the mortgage contract and the household's intertemporal budget constraint. Given an income process and the prevailing interest rate environment, each household dynamically decides how much to consume today and when to refinance its existing mortgage.

$$V(S_t, r_t) = \sup_{\{C_t, \rho_t^{refi}, \Delta F^{co}\}} \mathbb{E}_t \left[\int_0^\infty e^{-\delta t} \frac{C_t^{1-\gamma}}{1-\gamma} \, dt \right],$$

$$\text{s.t.} \quad dW_t = (Y_t - C_t + r_t W_t - m_t^* F_t)\, dt + \underbrace{dN_t^\chi \rho_t^{refi}}_{\text{refi decision}} \Big(\underbrace{\Delta F_t^{co}}_{\text{cash-out}} - \underbrace{\kappa}_{\text{refi fixed cost}} \Big),$$

$$dF_t = \underbrace{-\alpha F_t \, dt}_{\text{amortization}} + \underbrace{(F_t' - F_t) dN_t^\chi \rho_t^{refi}}_{\text{refi jump}},$$

$$dm_t^* = \underbrace{(m_t - m_t^*)\big(dN_t^m + dN_t^\chi \rho_t^{refi}\big)}_{\text{coupon reset at refinancing (moving + refi)}}.$$

The value function $V(S_t, r_t)$ represents a household's maximal expected lifetime utility given its current state. The individual state is $S_t \equiv (W_t, F_t, m_t^*, s_t)$, where W_t denotes liquid wealth, F_t the outstanding mortgage balance, m_t^* the existing mortgage coupon rate, and s_t the idiosyncratic income state. The aggregate state of the economy is summarized by the short rate r_t. The vector $\boldsymbol{\chi}$ collects the parameters governing refinancing frictions, including the attention and execution channels introduced above.

Households choose consumption C_t and refinancing-related actions, including whether to execute refinancing and the size of cash-out refinancing ΔF_t^{co}, to maximize discounted expected utility over an infinite horizon. Preferences are represented by a CRRA utility function with risk aversion parameter γ, and future utility is discounted at rate δ.

Refinancing opportunities arrive stochastically. In particular, households receive opportunities to consider refinancing according to a Poisson process dN_t^χ with intensity $\chi(gap_t, s_t)$, where the intensity depends on the refinancing incentive and the current income state. Conditional on such an arrival, households decide whether to execute refinancing subject to the execution (hassle-cost) threshold. In addition, exogenous events such as relocation or other forced mortgage resets trigger refinancing according to an independent Poisson process dN_t^m. These arrival processes jointly govern the timing of mortgage refinancing and coupon resets in the model.

It is important to emphasize that income affects refinancing behavior in a *non-monotonic* manner, even absent any mechanical targeting of refinancing frictions. On the one hand, lower-income households are more likely to be constrained borrowers. They face tighter payment-to-income (PTI) constraints and may lack the liquidity required to cover fixed refinancing costs. As a result, even when refinancing opportunities arise, these constraints limit their ability to act, giving rise to a *constraint channel* that lowers refinancing probabilities among low-income households.

On the other hand, lower-income households exhibit a strong *liquidity motive*. Because they place a higher value on immediate liquidity, they have stronger incentives to extract home equity through cash-out refinancing. Conditional on being able to refinance, they therefore tend to refinance more aggressively. The overall effect of income on refinancing and cash-out policy functions is thus non-monotonic, reflecting the interaction between these two opposing forces rather than a mechanical consequence of the model's frictional structure.

4.4 Uncertainty Environment

The model features two sources of uncertainty. First, households face idiosyncratic labor income risk, following the AR(1) process:

$$d \ln Y_t = \eta_y \, dt + \sigma_y \, dZ_t^y \tag{8}$$

Each household's labor income Y_i evolves stochastically according to a mean-reverting process with persistence parameter η_y and volatility σ_y. The innovation is driven by a standard Brownian motion Z_t^r that is independent across households and orthogonal to all aggregate shocks. This idiosyncratic income component introduces cross-sectional heterogeneity even in the absence of aggregate fluctuations.

Second, the aggregate short-term interest rate r_t follows a continuous-time stochastic process of the Cox–Ingersoll–Ross (CIR):

$$dr_t = -\eta_r(r_t - \bar{r})\,dt + \sigma_r\sqrt{r_t}\,dZ_t^r, \tag{9}$$

It is characterized by persistence η_r, volatility σ_r, and long-run mean $\bar{r}$. The process is driven by a common Brownian motion Z_t^r, which captures aggregate shocks affecting all households simultaneously. Changes in r_t propagate through to mortgage rates and refinancing incentives, thereby linking aggregate monetary conditions to household-level refinancing dynamics.

Together, the idiosyncratic income process (Y_i) and the aggregate short-rate process (r_t) generate both individual and aggregate uncertainty in the model. These shocks jointly shape the evolution of consumption, wealth accumulation, and mortgage refinancing behavior across the household distribution.

4.5 Mortgage Market Equilibrium

Each mortgage is modeled as a fixed-rate loan contract characterized by its coupon rate m and outstanding balance F. Instead of specifying a detailed amortization schedule, the model assumes that a constant fraction α of the loan is repaid each year. This simplification implies that repayments accelerate slightly as the loan approaches maturity, but it greatly reduces computational complexity without affecting the key results. When a household refinances, the coupon is replaced by a new rate m_t^* drawn from the prevailing market rate.

The mortgage price satisfies a risk-neutral pricing condition that accounts for the expected prepayment probability. Lenders fund themselves at the short rate and earn expected returns from mortgage payments such that, in equilibrium, they break even. In other words, the present value of a newly issued mortgage equals its face value, and the equilibrium coupon $m(r_t)$.

A key assumption is that lenders do not observe each household's refinancing friction. Instead, they form expectations based on the average refinancing hazard across the population ($\bar{\chi}$).[12]

$$P(W, F, r, m^*, s; \bar{\chi}) = \mathbb{E}\left[\int_0^{\tau} e^{-\int_0^t r_d \, du} \, m^* \, dt + e^{-\int_0^{\tau} r_d \, du} \,\middle|\, r_0 = r \right] \tag{10}$$

Under the assumption of complete markets, the equilibrium mortgage rate is determined such that the lender's expected profit is zero after accounting for anticipated refinancing behavior.

$$P(W, F, r, m^*, s; \bar{\chi}) = 1 \tag{11}$$

4.6 Dynamic Programming and Distributional Dynamics

Hamilton–Jacobi–Bellman Equation Given the continuous-time formulation and the presence of stochastic refinancing opportunities, the household's optimization problem admits a Hamilton–Jacobi–Bellman (HJB) representation. Let $V(S, r)$ denote the value function, where the individual state is $S = (W, F, m^*, s)$ and the aggregate state is summa-

[12] As noted by Berger et al. (2024), households with higher attention parameters are overrepresented among those who actually prepay, relative to the ex-ante distribution. This discrepancy between ex-ante and ex-post distributions can affect the lender's perceived hazard rate and, consequently, mortgage pricing. In the current version of the model, this feedback is not yet incorporated but will be addressed in a later refinement of the framework.

rized by the short rate r.

$$\delta V(r,S) = \sup_{C\geq 0} \Big\{ u(C) + \mathcal{L}_r V(r,S) + \mathcal{L}_y V(r,S) + \mu_W(r,S;C)\, V_W(r,S) + \mu_F(S)\, V_F(r,S) + \lambda^{\chi}(r,S) \left[\widetilde{V}^{refi}(r,S) - \Phi(s) - V(r,S)\right]_+ + \nu_m \left[V^m(r,S) - V(r,S)\right]\Big\}. \tag{12}$$

Equation (12) is a continuous-time HJB in which the first line captures standard flow utility and the endogenous savings decision. Consumption C is chosen optimally subject to the household's intertemporal budget constraint implied by the wealth drift μ_W. The derivatives V_W and V_F denote marginal values with respect to liquid wealth W and mortgage balance F, respectively, and $\mathcal{L}_r$ and $\mathcal{L}_y$ capture the effects of aggregate interest-rate dynamics and idiosyncratic income risk on continuation values.

The second line reflects endogenous refinancing opportunities. Refinancing consideration arrives according to a Poisson process with intensity $\lambda^{\chi}(r,S) \equiv \chi(\text{gap}, s)$, where gap $\equiv m^* - m(r)$ denotes the refinancing incentive. Upon such an arrival, households compare the continuation value from refinancing to the value of remaining in the current mortgage contract; the execution decision is governed by a hassle-cost threshold, so refinancing occurs only when the value gain exceeds the income-state–dependent threshold $\Phi(s)$.

The final term captures exogenous refinancing events, such as relocation, which arrive with constant intensity ν_m and mechanically reset the mortgage coupon to the prevailing market rate.

The cash-on-hand drift and mortgage-balance drift are

$$\mu_W(r,S;C) = Y(y) - C + rW - m^*F - \alpha F, \qquad \mu_F(S) = -\alpha F. \tag{13}$$

The coupon-reset (exogenous refinancing) jump value is

$$V^m(r,S) = V\big(r,\ (W,F,m(r),y)\big)\,, \tag{14}$$

and the value of refinancing with optimal cash-out is

$$\widetilde{V}^{refi}(r,S) \equiv \sup_{\Delta F^{co} \in \mathcal{D}(r,S)} V^{refi}\big(r,\ (W + \Delta F^{co} - \kappa,\ F + \Delta F^{co},\ m(r),\ y)\big)\,. \tag{15}$$

The infinitesimal generators for the short rate r (CIR) and idiosyncratic income y are

$$\mathcal{L}_r V \equiv \kappa_r(\bar{r} - r)\, V_r + \tfrac{1}{2}\sigma_r^2\, r\, V_{rr}, \qquad \mathcal{L}_y V \equiv \mu_y(y)\, V_y + \tfrac{1}{2}\sigma_y^2\, V_{yy}. \tag{16}$$

Distribution Dynamics Let $g_t(r,S)$ denote the cross-sectional density of households over individual states $S = (W, F, m^*, y)$ and the aggregate short rate r. The evolution of g_t is governed by a Kolmogorov Forward Equation (KFE) that combines continuous drift in (W, F, r, y) with jump transitions generated by refinancing and coupon resets. Define the effective endogenous refinancing intensity as $\lambda^{refi}(r,S) \equiv \lambda^{\chi}(r,S)\,\rho(r,S), \qquad \rho(r,S) \in \{0,1\}$, where $\rho(r,S)$ is the household's refinancing policy (equal to one when the value gain exceeds the threshold and the household refinances upon an attention arrival). Then $g_t(r,S)$ satisfies

$$\begin{aligned}
\partial_t g_t(r,S) \;=\; & -\partial_W\big(\mu_W(r,S), g_t(r,S)\big) - \partial_F\big(\mu_F(S), g_t(r,S)\big) + \mathcal{L}_r g_t(r,S) + \mathcal{L}_y g_t(r,S) \\
& - \big(\nu_m + \lambda^{refi}(r,S)\big) g_t(r,S) + \nu_m g_t\big(r, (\mathcal{T}^m)^{-1}(S)\big) \\
& + \lambda^{refi}\big(r, (\mathcal{T}^{refi})^{-1}(S)\big) g_t\big(r, (\mathcal{T}^{refi})^{-1}(S)\big)
\end{aligned} \tag{17}$$

where $\mu_W(r,S) = Y(y) - C(r,S) + rW - m^*F - \alpha F$ and $\mu_F(S) = -\alpha F$ are the endogenous drifts in liquid wealth and mortgage balance. The operators $\mathcal{L}_r^*$ and $\mathcal{L}_y^*$ are the adjoints of the r- and y-generators and capture diffusion-driven mass transport in those dimensions:

$$\mathcal{L}_r^* g \equiv -\partial_r\big(\kappa_r(\bar{r} - r)\, g\big) + \tfrac{1}{2}\,\partial_{rr}\big(\sigma_r^2\, r\, g\big), \qquad \mathcal{L}_y^* g \equiv -\partial_y\big(\mu_y(y)\, g\big) + \tfrac{1}{2}\,\partial_{yy}\big(\sigma_y^2\, g\big) \tag{18}$$

The first line of (17) captures continuous evolution of the distribution due to saving/consumption choices, amortization, and the stochastic dynamics of r and y. The second line implements jump transitions. Mass leaves state (r,S) at total intensity $\nu_m + \lambda^{refi}(r,S)$ (outflow), and enters from pre-jump states that map into S through the coupon-reset mapping $\mathcal{T}^m$ and the refinancing mapping $\mathcal{T}^{refi}$ (inflow). In practice, for the mappings used in the numerical implementation, the Jacobian terms are equal to one and are omitted.

In computation, I solve the HJB and KFE jointly using an implicit finite-difference scheme with up-winding for the drift terms and sparse transition matrices for the jump components.

5 Quantitative Analysis

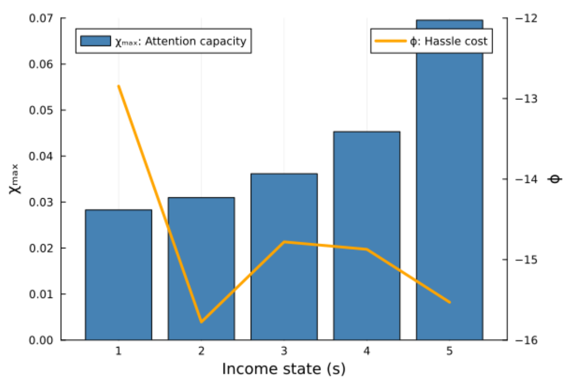
5.1 Calibration

Attention Parameters The attention friction is governed by the parameters $\chi_{\max}(s)$, k, and μ. In the baseline calibration, I treat $\chi_{\max}(s)$ and the income-specific hassle-cost thresholds $\Phi(s)$ as a joint block of ten friction parameters and estimate them simultaneously from the refinancing hazard by income quintile. This approach allows the data to determine how much of the cross-sectional variation in hazards is absorbed by attention capacity versus execution costs, rather than imposing a priori structure on either channel.

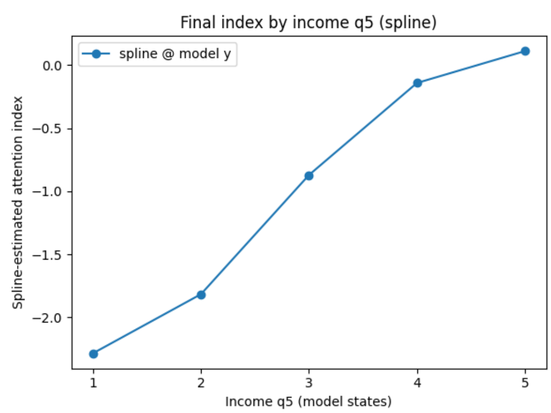
The estimated attention capacities display a clear income gradient. The function $\chi_{\max}(s)$ rises monotonically with the income state, implying that higher-income households have a higher ceiling for how often they pay attention to refinancing opportunities when they are deep in-the-money. To provide external evidence for this pattern, I use household-level proxy measures constructed from the National Survey of Mortgage Originations (NSMO). The survey reports (i) the number of lenders a borrower considered and

(ii) the number of lenders to which the borrower applied at origination. These variables capture the intensity of search and information acquisition during mortgage choice. I normalize the responses and aggregate them into a single attention index by income group. Figure 5 compares this NSMO-based index with the model-implied $\chi_{\max}(s)$ and shows that both series increase systematically with income, lending support to the interpretation of $\chi_{\max}(s)$ as an income-dependent attention capacity.

By contrast, the estimated hassle-cost thresholds $\Phi(s)$ are not monotone in income. In particular, the second income quintile exhibits a markedly lower threshold than both the bottom and middle groups. Figure 5a plots the estimated $\Phi(s)$ across quintiles. A useful way to interpret this pattern is that households in the lower-middle of the income distribution sit at the intersection of two opposing forces. On the one hand, they remain relatively liquidity constrained and have strong incentives to refinance and extract equity when opportunities arise. On the other hand, they must still cover the upfront refinancing cost κ, which limits refinancing for the very lowest-income borrowers. A lower effective hassle cost for the second quintile helps the model reconcile these forces and match the observed refinancing behavior in this part of the distribution.



(a) Estimated friction parameters



(b) NSMO-based attention index (normalized relative to quintile 3)

Figure 5: Friction parameters and NSMO attention index by income quintile

Gap Sensitivity Parameters The remaining attention parameters, k and μ, govern how attention intensity responds to the refinancing incentive, measured by the interest rate gap. The parameter μ controls the location of the attention response, while k determines the sensitivity of attention intensity to changes in the gap. I restrict these parameters to be common across income states.

This restriction is motivated by reduced-form evidence from the data. Across income groups, the increase in refinancing activity as the interest rate gap widens exhibits similar curvature and timing, with differences across groups primarily reflected in the maximum refinancing rates rather than in the responsiveness to the gap itself. Allowing $\chi_{\max}(s)$ to vary by income state while keeping k and μ common is therefore sufficient to capture the observed heterogeneity in refinancing behavior.

I estimate k and μ using Nonlinear Least Squares (NLS). Specifically, for each income quintile, I fit the gap-dependent attention specification implied by equation (7) to the empirical refinancing hazard as a function of the interest rate gap. The estimation jointly disciplines the slope and location of the attention response while allowing the level to differ across income groups through $\chi_{\max}(s)$. This approach ensures that the calibrated attention function matches the observed shape of refinancing responses to interest rate incentives without overfitting income-specific sensitivities.

Other model parameters The other model parameters follows the quantitative macro-finance literature. The coefficient of relative risk aversion is set to $\gamma = 2$, a standard value in macroeconomic models. The baseline mortgage balance F, the time-preference parameter δ, and the parameters governing the interest rate process — $(\eta_r, \sigma_r, \bar{r})$ — are calibrated to match Berger et al. (2021) , implying a mortgage balance of \$150,000, an annual mean short rate of 3.5%, and an interest-rate volatility of 6%. The stochastic income process follows Floden and Lindé (2001), with a log-income persistence corresponding to a half-life of 7.3 years and an annualized volatility of 21%.

For refinancing frictions, two types of behavioral parameters, $\chi_1(s)$ and $\chi_2(s)$, capture the heterogeneous intensity of refinancing attention and the execution delay across income quintiles, respectively. Both parameters are modeled as five-element vectors corresponding to income quintiles $s = 1, \ldots, 5$.

The payment-to-income ratio $\phi^{PTI} = 0.43$ limits the maximum feasible mortgage payment relative to income, while the loan-to-value ratio $\phi^{LTV} = 0.8$ restricts the maximum outstanding balance relative to home value. Both parameters are calibrated to match average underwriting standards in the Freddie Mac.

Further details on the cross-sectional distribution of the baseline model are provided in Figure B.5.

Table 4: Model Parameter Values

Panel A: Exogenous Processes			
Parameter	**Value**	**Description**	**Source**
$\ln 2/\eta_r$	5.3 years	Half-life of interest-rate shock	Berger et al. (2021)
$\bar{r}$	3.5% (p.a.)	Unconditional mean short rate	Berger et al. (2021)
σ_r	6% (p.a.)	Volatility of short rate	Berger et al. (2021)
$\ln 2/\eta_y$	7.3 years	Half-life of (log) income shock	Flodén and Lindé (2001)
$\mathbb{E}[Y_t]$	\$58,000	Unconditional mean income	Flodén and Lindé (2001)
σ_y	21% (p.a.)	Log-income volatility	Flodén and Lindé (2001)
γ	2	Inverse of IES	Macro literature standard
F	\$150,000	Mortgage balance outstanding	Berger et al. (2021)

Panel B: Refinancing Frictions			
Parameter	**Hybrid**	**Description**	**Source**
ν_m	4.1% (p.a.)	Arrival rate of exogenous moving shocks	Berger et al. (2021)
$\chi_{\max}(s)$	{0.028, 0.031, 0.036, 0.045, 0.069}	Maximum attention capacity by income quintile	Estimated
$\log \Phi(s)$	{−12.8, −15.8, −14.8, −14.8, −15.5}	Maximum attention capacity by income quintile	Estimated
k	1.931	Responsiveness of attention to rate gap	Estimated
μ	0.284	Location parameter of attention response to rate gap	Estimated
κ	\$3,000	Fixed (menu) cost of refinancing	Eichenbaum et al. (2021)
ϕ^{PTI}	0.43	Payment-to-income constraint ratio	Freddie Mac
ϕ^{LTV}	0.80	Maximum loan-to-value ratio	Freddie Mac

5.2 Hazard Rate

I now turn to evaluating the performance of the model. As a first step, I examine how well it replicates the targeted moments—refinancing hazard rates by rate-gap bins across income quintiles. The exercise is conducted by feeding the market coupon distribution and the realized short-rate path as of January 2005 into the model[13], and comparing the model-generated outcomes with the empirical estimates from the data.

The goal is to assess how well the model reproduces the heterogeneous refinancing patterns documented in the data. In particular, the calibration targets two key dimensions of refinancing behavior: *intensity* and *timing*. Refinancing intensity captures how sensitively households respond once they become in-the-money—driven mainly by the attention parameter, χ_1. Refinancing timing captures how quickly households act after recognizing the benefit. Even when refinancing is clearly optimal, many households delay execution—a friction captured by χ_2.

Figure 6 shows the model fit against the data. Although the model is not yet fully calibrated, it replicates several salient empirical patterns. The baseline model (blue dashed) reproduces the higher refinancing peaks observed among the fourth and fifth income quintiles compared to the bottom groups.[14] By contrast, when income-specific heterogeneity is suppressed and homogeneous frictions are imposed, the model (orange dotted) tends to overpredict refinancing among low-income borrowers and underpredict it among high-income borrowers, generating systematic distortions relative to the data.

I next present a simple aggregate diagnostic of model fit: the average monthly refinancing probability among *in-the-money* borrowers, plotted by income quintile. Figure 7 shows that this cross-income pattern is informative precisely because it is not mechanically implied by standard benchmark setups. In the data, refinancing propensity increases monotonically with income, producing an upward-sloping profile across quintiles.

[13]The empirical sample begins in 2000, but the model simulation starts from the mid-sample to allow for a sufficient burn-in period.

[14]The current calibration is preliminary and will be refined further.

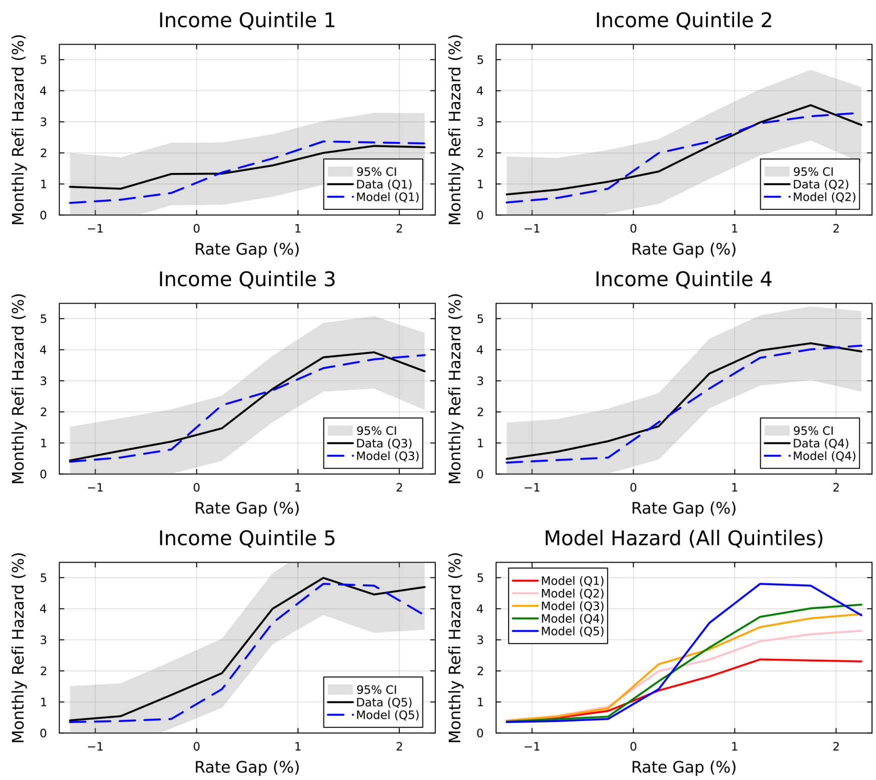
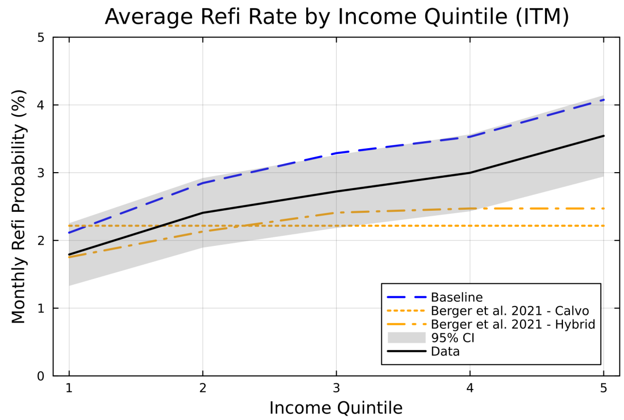


Figure 6: Hazard Rate by Gap Bin: Model vs. Data

The baseline model with income-dependent frictions reproduces this upward-sloping profile well, even if it somewhat overpredicts the overall level of refinancing relative to the data. In contrast, a Calvo-style attention-only benchmark with common parameters across households generates an essentially flat profile by construction. A hybrid benchmark that allows for additional channels beyond pure attention produces only a muted gradient, reflecting the absence of an income-dependent execution wedge. Taken together, these comparisons highlight that matching aggregate refinancing hazards is not sufficient: without heterogeneity in refinancing frictions across income groups, the model cannot replicate the inequality in refinancing propensity that motivates the paper and underlies heterogeneous monetary transmission.



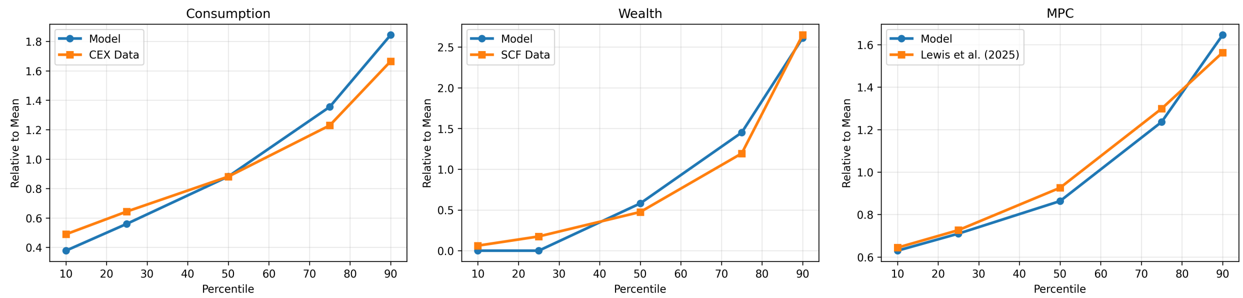
(a) Baseline vs Berger et al. (2021)

Figure 7: Average Hazard Rate by Income Quintile: Model vs. Data

5.3 Untargeted Moments

Figure 8 reports untargeted distributional moments in the baseline model. I compare the Lorenz curves for consumption and liquid wealth to their empirical counterparts from the CEX (1988–2018) and the 2001 SCF, and I contrast the model-implied MPC profile with the estimates in Lewis et al. (2025). All objects are computed at percentiles of the cross-sectional distribution and normalized by their means, so the comparison emphasizes shapes rather than levels. The model reproduces the broad distributional patterns in the data—realistic inequality in liquid wealth and consumption—and generates an MPC profile that is close to the empirical benchmark.

This validation is important for interpreting the paper's transmission results. In the model, refinancing affects consumption through a cash-flow channel, and the magnitude of this response depends on households' marginal propensities to consume, which are tightly linked to liquid-wealth positions. It is therefore reassuring that, even though these moments are not directly targeted, the calibrated environment delivers plausible liquid-wealth inequality and MPC heterogeneity, reducing the concern that the consumption IRFs and counterfactual transmission results are driven by an unrealistic liquidity or MPC distribution.



(a) Consumption (b) Liquid Wealth (c) MPC

Figure 8: Lorenz Curve and MPC: Model vs. Data

The Figure 9, compares the evolution of average mortgage coupon rates in the model to the data. In the empirical series I focus on outstanding mortgages, exclude purchase originations, and compute the average coupon rate over the remaining loans. The model counterpart is constructed in the same way, based on the simulated cross-sectional distribution of contracts. For reference, I also plot the 30-year fixed mortgage rate on new originations, which does not coincide with the average coupon on outstanding debt but traces the underlying interest rate path. The model starts from a similar initial level and tracks the subsequent decline in the average coupon rate, although the adjustment is somewhat slower. This residual gap appears to be related to differences between the model-implied distribution of borrowers across income groups and the distribution observed in the data.

5.4 Counterfactual Experiments: Homogeneous Frictions

To assess whether the model can explain refinancing inequality without heterogeneity in refinancing frictions, I consider a counterfactual in which frictions are imposed uniformly across income groups. Specifically, I set the attention-arrival parameter and the execution wedge to be common across income states—$\chi_{\max}(s) = \bar{\chi}$ and $\Phi(s) = \bar{\Phi}$ for all s—and recalibrate the common parameters to match overall refinancing activity. Figure 10 shows that this homogeneous-frictions specification can match some aspects of the average level

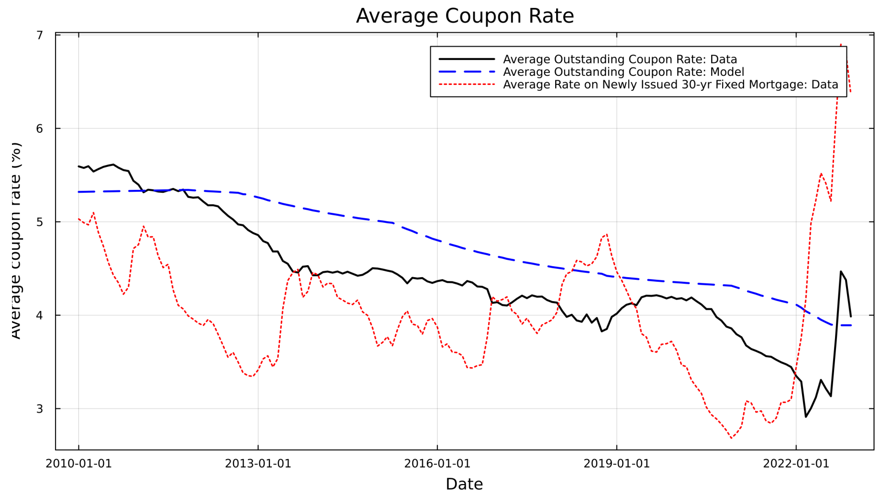


Figure 9: Average Coupon of Outstanding Loans: Model vs. Data

of refinancing, but it fails on the central object of interest: the systematic cross-income differences in refinancing behavior.

In particular, the homogeneous-frictions model cannot reproduce the income gradient in refinancing. Relative to the baseline model, it delivers hazard profiles that are too similar across income quintiles and an average in-the-money refinancing rate that is nearly flat by income, in contrast to the sharply upward-sloping pattern observed in the data. This comparison provides a key source of discipline for the model: explaining refinancing inequality requires income-dependent frictions. Put differently, heterogeneity in refinancing frictions is not a second-order detail; it is essential for matching the joint distribution of refinancing propensities across income groups and, consequently, for capturing heterogeneous monetary transmission.

5.5 Counterfactual Experiments: Friction Channels and Heterogeneity

To assess how much each refinancing friction and its income heterogeneity contribute to the model fit, I run a set of counterfactual experiments that selectively remove individual channels. In the first experiment, shown in Figure 11, I shut down the hassle-cost channel and retain only the gap-dependent attention friction, recalibrating the remaining five

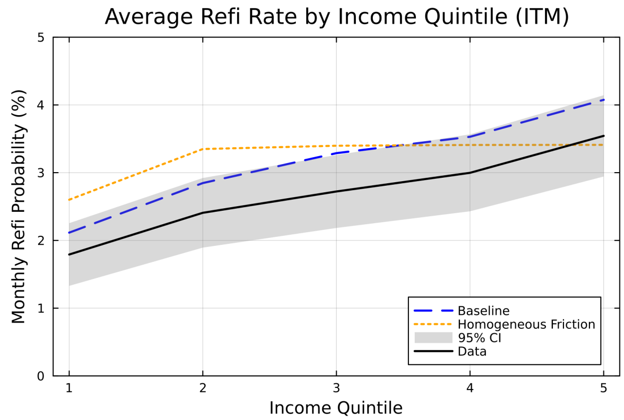


Figure 10: Average Hazard Rate: Counterfactual Homogeneous Frictions

attention parameters by income group. The hazard curves for the bottom four quintiles remain close to the baseline, but the model now understates refinancing activity for the top quintile, especially at larger rate gaps. The intuition is that removing the execution threshold raises the continuation value for high-income households that expect to move down the income distribution in the future. Because refinancing becomes more attractive in lower income states, the expected gain from refinancing while still in quintile 5 falls, which depresses current hazards.

Figure 12 instead shuts down attention frictions and lets refinancing be determined entirely by the policy function implied by hassle costs. In this case the model resembles a standard menu-cost environment: refinancing probabilities are near zero for small gaps and jump steeply once the threshold is crossed. This specification cannot match the empirical feature that refinancing rates remain well below one even for deep in-the-money borrowers. Figure 13 complements this result by plotting the average refinancing rate among in-the-money borrowers across income quintiles in the hassle-cost-only specification. The profile is almost flat: apart from a modest decline in the first quintile driven by tighter liquidity and payment-to-income constraints, refinancing rates are essentially identical across the remaining groups, much like in the pure Calvo model of Berger et al. (2021). Hence a pure menu-cost formulation cannot generate the refinancing inequality observed in the data.

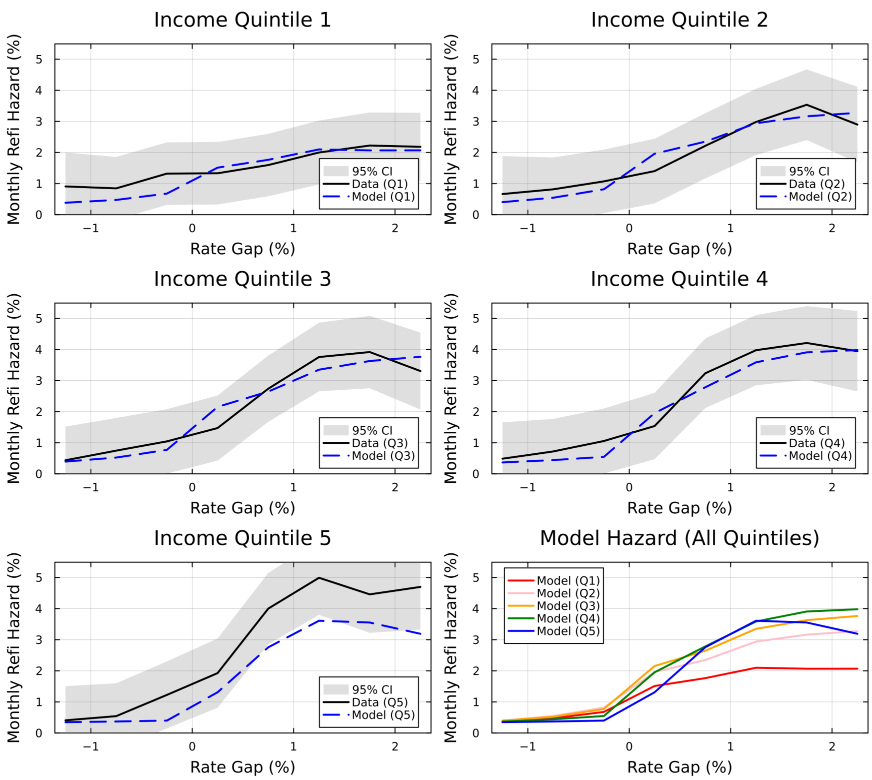


Figure 11: Hazard Rate by Gap Bin: Attention Friction Only

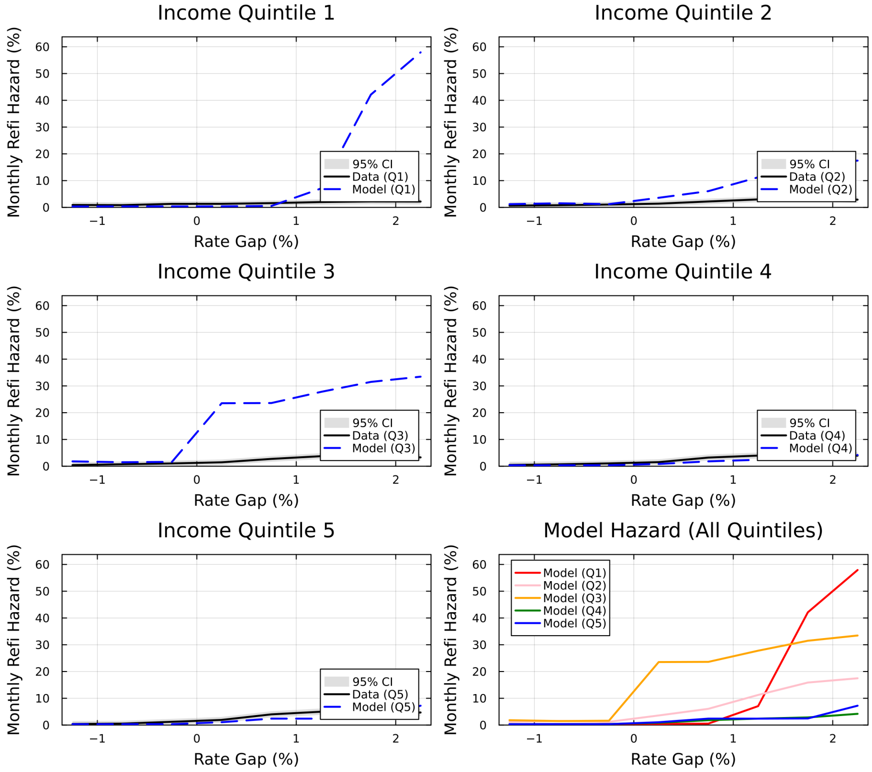


Figure 12: Hazard Rate by Gap Bin: Hassle Cost Only

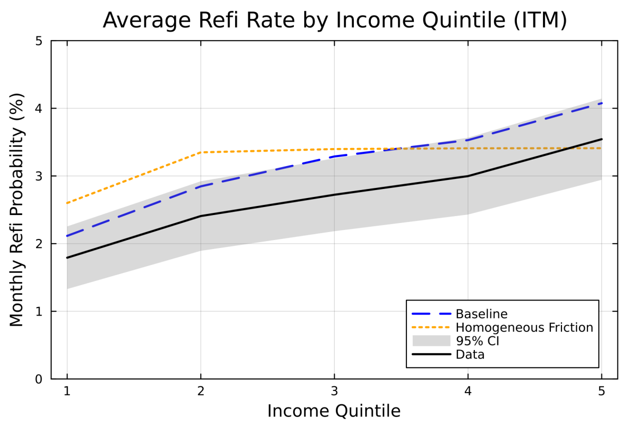


Figure 13: Average Hazard Rate by Income Quintile: Hassle Cost Only

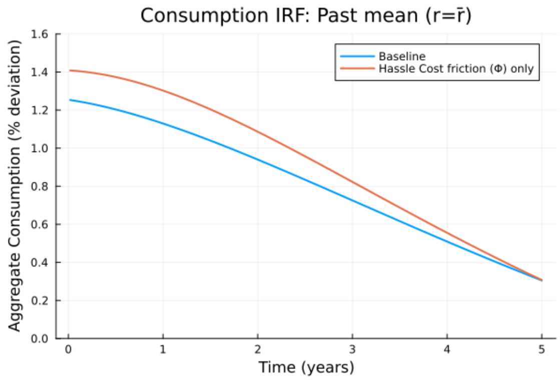
5.6 Consumption Response to Monetary Policy

This appendix reports a supplementary quantitative exercise that links the paper's refinancing-inequality facts to a standard mechanism for monetary transmission. When interest rates fall, some borrowers become *in-the-money*, yet refinancing frictions may prevent them from refinancing. If refinancing is delayed or forgone, the cash-flow relief from lower mortgage payments reaches fewer households—and arrives later—thereby weakening the expansionary effects of monetary policy.

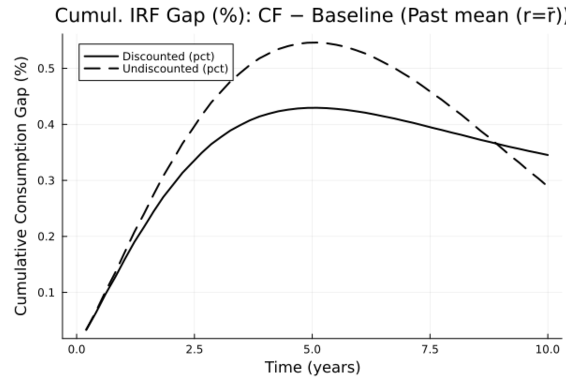
The ultimate goal of this paper is to assess how refinancing inequality affects the transmission of monetary policy. When interest rates fall, some borrowers become *in-the-money*, yet frictions may prevent them from refinancing. If refinancing is delayed or forgone, the expansionary effects of monetary policy are weakened.

To quantify this mechanism, I compare the consumption response to a 100-basis-point decline in the policy rate across different frictional environments. The main experiment considers a "reduced-frictions" counterfactual in which refinancing frictions are equalized across income groups at the level faced by the top income quintile in the baseline model. Concretely, I set the friction parameters for all income states to $\chi_{\max}(s) = \chi_{\max}(5)$ and $\Phi(s) = \Phi(5)$, holding the remaining elements of the environment fixed.

Figure 14 illustrates the consumption responses to a 100 basis-point interest rate cut under the baseline and the counterfactual scenario with higher χ_{max}. In the counterfactual, all income groups are assumed to have the same level of attention as the top quintile in the baseline model. As shown in the left panel, the counterfactual produces a stronger and more persistent consumption response starting immediately after the policy shock. The right panel plots the cumulative difference between the two paths. By two and a half years after the shock, the counterfactual implies an additional rise in present-value consumption of about 0.43%p after five years. Given that the baseline model produces a five-year cumulative response of 3.13%p, this counterfactual represents an amplification of about 13.7%.



(a) Consumption IRF



(b) Cumulative Consumption IRF

Figure 14: Consumption Response to a 1 pp Rate Cut

To understand where this amplification comes from, I decompose the five-year cumulative (discounted) consumption response by income quintile. Figure 15 shows that the additional present-value consumption generated by the counterfactual is concentrated in the lower part of the income distribution, with the largest contribution coming from quintiles 1 and 2. This pattern is consistent with the model's MPC heterogeneity: payment relief delivered through refinancing has a larger effect on aggregate consumption when it accrues to high-MPC households, which are disproportionately represented among lower-income borrowers.

A key point to note is that this counterfactual is not purely hypothetical. Byrne et al. (2023) provide evidence from a field experiment that a simple refinancing notice can substantially raise refinancing attention—by roughly 60 percent. This suggests that policy or institutional interventions that increase households' effective attention capacity, the primary driver of the in-the-money refinancing friction captured by $\chi_{\max}$, are feasible in practice. If such interventions disproportionately relax frictions for borrowers who are most constrained in refinancing, they would both narrow the cross-income gap in refinancing outcomes and strengthen monetary transmission. Taken together, these results highlight a key quantitative implication of refinancing inequality for monetary policy: reducing frictions precisely where refinancing is most constrained not only compresses inequality in refinancing outcomes, but also raises the aggregate potency of monetary policy by reallocating interest-payment savings toward households with higher marginal propensities to consume.

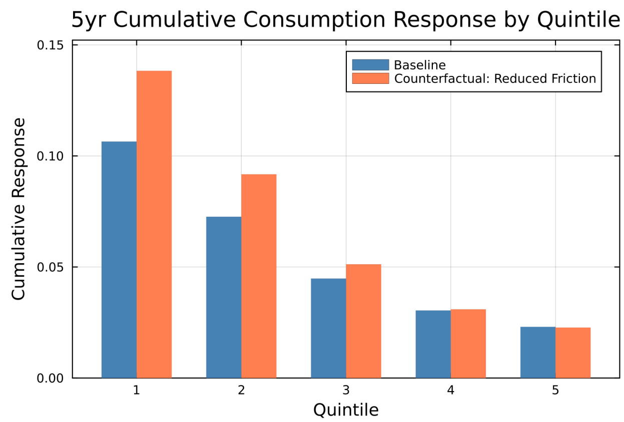


Figure 15: Consumption Response Decomposition: Baseline vs. Reduced Frictions

5.7 Implications on Inequality

Finally, I examine how the reduced-frictions counterfactual changes long-run inequality in the stationary distribution, rather than along the transition path following a monetary shock. Table 5 compares inequality measures under the baseline calibration and the reduced-frictions economy. The results show little movement in consumption inequality: the consumption Gini remains at 0.30 in both economies (a change of 0.00), the top-10 consumption share is essentially unchanged at 0.21%, and the bottom-50 consumption share is also stable at 0.28%.

In contrast, liquid-wealth inequality improves meaningfully. The liquid-wealth Gini declines from 0.56 to 0.53 (a reduction of 0.03, or 5.47%), and the top-10 liquid-wealth share falls from 0.33% to 0.31% (a decline of 0.02 percentage points, or 4.82%). At the same time, the bottom-50 liquid-wealth share rises from 0.09% to 0.11% (an increase of 0.02 percentage points, or 25.47%). Taken together, the stationary comparisons indicate that reducing refinancing frictions has limited effects on long-run consumption inequality but leads to a notable compression of liquid-wealth inequality, consistent with refinancing easing liquidity constraints and shifting the cross-sectional distribution of liquid assets.

Table 5: Stationary Inequality under Reduced Refinancing Frictions

	Consumption				Liquid wealth			
Metric	Base	CF	CF–Base	%Δ	Base	CF	CF–Base	%Δ
Gini	0.30	0.30	0.00	0.44%	0.56	0.53	-0.03	-5.47%
Top 10 share	0.21%	0.21%	0.00%p	0.01%	0.33%	0.31%	-0.02%p	-4.82%
Bottom 50 share	0.28%	0.28%	0.00%p	-0.37%	0.09%	0.11%	0.02%p	25.47%

6 Conclusion

This paper documents that refinancing inequality — the systematic gap in refinancing propensities across income groups — is large, persistent, and consequential for monetary policy transmission. Using a broader time series and more robust measures than previous studies, I show that low-income households refinance not only less but also later than high-income households, even when refinancing is clearly profitable. The implied unrealized savings are economically meaningful and are concentrated precisely among lower-income borrowers who stand to benefit most. To account for these patterns, I develop a mortgage refinancing model with two distinct and income-dependent sources of inertia: inattention to refinancing opportunities ($\chi_{max}(s)$), which explains why hazard rates remain well below one even deep in the money, and execution and hassle costs ($\Phi(s)$), which shape the delays in acting on profitable opportunities. A model with homogeneous frictions fails to reproduce the observed income gradient; both frictions are required to match the data.

The quantitative implications are substantial. Reducing refinancing frictions for low-income borrowers raises the five-year discounted cumulative consumption response to an interest rate cut by approximately 13.7%. Notably, the inequality effects are concentrated in liquid wealth — the bottom-50 share of liquid wealth rises meaningfully — while consumption inequality itself moves relatively little. These findings highlight refinancing inequality as a significant attenuator of monetary easing, and point to friction-reducing interventions targeted at low-income borrowers as a channel through which the real effects of monetary policy could be meaningfully amplified.

A Appendix: Refinancing Response to Monetary Policy

While the main text focuses on documenting refinancing inequality and its empirical determinants, this appendix provides additional evidence on a natural quantitative implication: how such inequality can shape the transmission of monetary policy through the refinancing channel. When interest rates fall, some borrowers become *in-the-money*, yet refinancing frictions may prevent them from responding promptly. If refinancing is delayed or forgone—especially among households with higher marginal propensities to consume—the expansionary effects of monetary policy through mortgage-payment relief are correspondingly weakened.

A.1 Data

To demonstrate refinancing inequality in a dynamic context, I examined the refinancing responses to exogenous monetary shocks across income quintiles. To do this, I followed the methodology of Eichenbaum et al. (2022) and used Freddie Mac panel data converted into regional data. The advantage of using regional aggregate data is twofold: first, it allows for the comparison and verification of results against established research; more importantly, it enables the use of the entire Freddie Mac population data rather than being limited to the old-new refinancing matched panel used in the previous analysis[15]. While the earlier panel used loan-level data to meticulously control for loan and borrower characteristics, this analysis focuses on refinancing responses in a macroeconomic context, making regional aggregate data appropriate. I constructed a regional panel using Freddie Mac data, consisting of quarterly numbers of outstanding and refinancing loans by income quintile for mortgages originated from 2000 to 2019 across 908 three-digit ZIP code areas[16].

[15]Although refinancing information (reasons for prepayment) is not available at the loan level in the Freddie Mac data, which required me to match old and new loans to ensure that the loan was prepaid due to refinancing, it can be observed in the origination data. By aggregating the issuance data by region, I can use the entire dataset to derive the share of refinances relative to total outstanding loans.

[16]However, the actual analysis used data from 2004 onwards. This is because Freddie Mac data provides information only on loans originated from 2000 onwards, leading to insufficient cumulative loan data by

For monetary policy shocks, I used the high-frequency shock data from Swanson (2021) to ensure exogeneity. Swanson recorded the changes in various asset prices within a 30-minute window around every FOMC meeting from 1991 to 2019. He then extracted the three largest principal components and demonstrated that these components correspond to changes in the federal funds rate, forward guidance, and large-scale asset purchases (LSAPs). A key advantage of this approach is that it addresses the limitations of high-frequency identification during the zero lower bound (ZLB) period. Since a significant portion of my data includes the ZLB period, Swanson's shocks are particularly relevant.

I obtain aggregate time-series variables, including forecasts of unemployment, inflation, and GDP, from the Survey of Professional Forecasters (SPF). These variables were used to control for the information channel suggested by Nakamura and Steinsson (2018) . Also, to control for house prices, I used the Zillow ZIP code-level house price index.

In this analysis, the key variable is the refinancing propensity. I compute this by measuring the cumulative proportion of refinanced loans from outstanding loans at the regional level as below:

$$\rho_{t+h}^{r,q} = \frac{x_t^{r,q} + \ldots + x_{t+h}^{r,q}}{(y_t^{r,q} + \ldots + y_{t+h}^{r,q})/h} \tag{19}$$

Here, $x_t^{r,q}$ is the number of refinancing loans by income quintile q in region r in quarter t, $y_t^{r,q}$ is the number of all outstanding loans, and h is the analysis time horizon.

One limitation is worth noting before proceeding. Although the underlying loan-level data cover originations from 2000 onward, I begin the regression sample in the second quarter of 2004 to allow for a sufficient burn-in period—that is, until the stock of outstanding loans becomes large enough. Given that the identified monetary policy shocks exists up to the second quarter of 2019 and that the regressions include lagged

income quintile up to 2003. Therefore, to prevent outlier values in income quintile refinancing rates, a sort of burn-in period was necessary.

variables, the effective time dimension consists of 56 quarterly observations. This relatively short time span makes clustering by time infeasible, as it would result in excessively large standard errors. At the same time, clustering at the monthly level is infeasible since the matched-sample structure does not provide sufficient observations per month. To address this small-T issue, I instead employ block bootstrapping by time, which corrects the standard errors for the limited number of time periods. In addition, following the recommendation of Almuzar and Sancibrian (2024), I include a sufficient number of lags (six quarters) to mitigate this concern.

A.2 Regression Results

Average Refinancing Response to Monetary Policy Shocks I begin by examining the average refinancing response for all income groups, in comparison with existing research. I start by considering the local projection analysis suggested by Jordà (2005).

$$\rho^r_{t+h} = \beta^h_0 X^r + \beta^h_1 \Delta R^{IV}_t + \beta^h_2 Z^r_t + \beta^h_3 Z_t + Lags^r_t + \epsilon^z_t \tag{20}$$

Here, X^r represents a vector of region fixed effects, and ΔR^{IV}_t denotes an instrument variable for interest rate *falls* using using high-frequency shocks. The interest rate is transformed to a positive sign indicating a decrease in interest rates. Therefore, I expect a positive response of refinancing. Z^r_t includes regional control variables such as changes in unemployment rates and house prices. The vector Z_t denotes a set of time-varying controls, specifically including a two-year ahead horizon, the civilian unemployment rate (also two years ahead), and the CPI inflation rate (forecasted for both one and two years ahead). I used three lags terms for all variables.

First, I show the analysis by demonstrating the strong impact of monetary policy on mortgage interest rates. Table 6 presents the first-stage estimates. All three high-

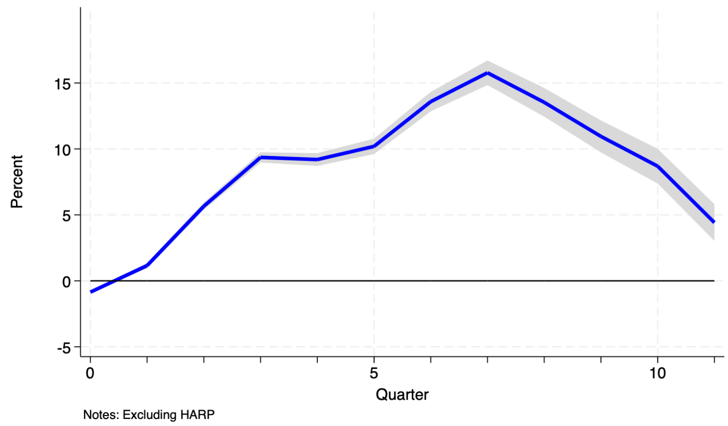


Figure 1: Cumulative Average Refinancing Response to a 1% Interest Rate Cut

Notes: This plot shows the cumulative refinancing response of all income groups to 1% interest rate falls using Equation 20. The dependent variable is the cumulative proportion of loans refinanced in each region as described in Equation 19. Regions are 908 three-digit ZIP code, and quarterly data from 2004 to 2022 was used. The shading represents the 95% confidence interval.

frequency factors exhibit a significant positive correlation with the 30-year fixed mortgage interest rate. In particular, the effect of LSAP, which involves the purchase of long-term bonds, has the largest impact on the 30-year fixed mortgage rate, aligning with expectations. The negative relationship between rising expectations for GDP and CPI and the actual mortgage rate reduction effect is consistent with economic interpretation, as higher expectations for GDP and CPI reduce the real impact of mortgage rate cuts. The F test for the joint significance of the regression coefficients is greater than ten.

Figure 1 shows the average refinancing response to a 1% interest rate falls. It indicates that cumulatively 15% of all mortgages are refinanced 7 quarters after the shock. After the seventh quarter, the cumulative ratio decreases because new mortgages increase the total number of outstanding loans while the number of refinancing decreases, gradually lowering the ratio below the unconditional ratio. Although the types of mortgages covered by the data and the sample periods differ from Eichenbaum et al. (2022)'s results, which

Table 6: First-stage Estimates

Dependent		Δ Real Mortgage Rates of 30-year FRM	
High-frequency Factors		Survey of Professional Forecasters	
- FFR	0.0165[a]	- ΔGDP (2 yrs)	-0.0341[a]
	(0.00)		(0.00)
- Forward Guidance	0.0290[a]	- Unemp. (2 yrs)	0.0273[a]
	(0.00)		(0.00)
- LSAP	0.0344[a]	- CPI (1 yrs)	-0.0347[a]
	(0.00)		(0.00)
ΔHouse Price	0.0005[a]	- CPI (2 yrs)	-0.0832[a]
	(0.00)		(0.01)
Constant	0.0427[a]	Unemp.	0.00641[a]
	(0.02)		(0.00)
Observations	46,351	*F*-statistic	2,803.820
R-squared	0.358	Prob. > F	0.000

Notes: Superscript a, b, c denote significance at 1%, 5%, 10% levels respectively.
S.E. in parenthesis.

show about 15-25%[17] refinancing by the fifth quarter following an interest rate shock, the result is consistent with their findings.

Heterogeneous Refinancing Response Across Income Groups Now, I will examine the heterogeneous refinancing responses across different income levels, which is the primary focus of this chapter. By modifying the Equation 20 I consider following specification:

$$\rho^{r,q}_{t+h} = \beta^h_0 X^r + \beta^h_1 \Delta R^{IV}_t + \beta^h_2 Z^r_t + \beta^h_3 Z_t + Lags^{r,q}_t + \epsilon^z_t \tag{21}$$

While the meaning of each variable remains the same as in Equation 20, the dependent variable, which is the proportion of refinanced loans, was analyzed separately for each income level. Figure 2 illustrates the refinancing response by income quintiles to a 1% interest rate reduction. Seven quarters after the interest rate reduction, households in the highest income quintile show a cumulative refinancing rate of approximately 15%. In contrast, households in the bottom income quintiles have peak cumulative refinancing rates of about 10%, significantly lower than their wealthier counterparts.

[17] They estimated the refinancing rate depending on the interest rate gaps between existing loans and the market rate, which makes the estimated range larger.

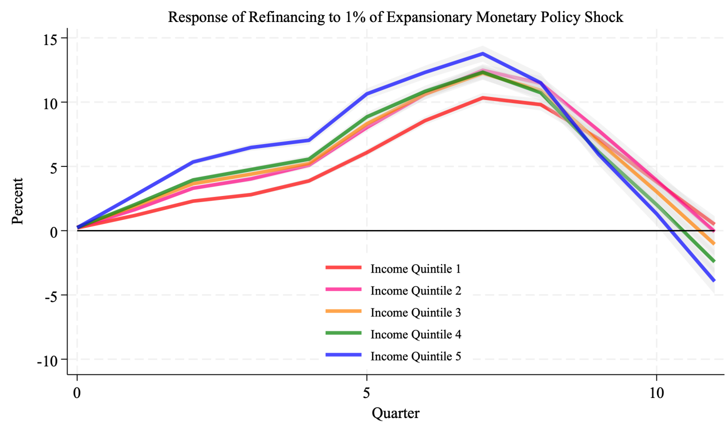


Figure 2: Cumulative Refinancing Response to a 1% Interest Rate Cut by Income Quintile

Notes: This plot shows the cumulative refinancing response by income quintile to 1% interest rate cuts using Equation 21. The dependent variable is the cumulative proportion of loans refinanced in each region and income group as described in Equation 19. Regions are 908 three-digit ZIP code, and quarterly data from 2004 to 2022 was used. The shading represents the 95% confidence interval.

There are other specifications to measure the dependent variable of the refinancing proportion. As explained in the Appendix subsection C.5, I conducted robustness checks using different specifications and the pattern of refinancing inequality was robust across all specifications.

B Appendix: Tables and Figures

B.1 Figures and Tables

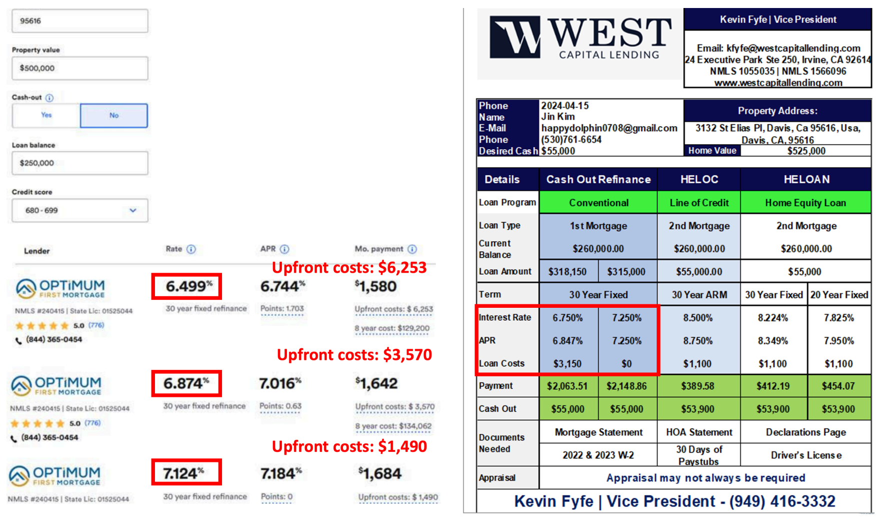


Figure B.3: Predicted Interest Rate Reduction from Refinancing

Notes: This figure shows the offers received from mortgage brokers in a hypothetical scenario for refinancing a home in Davis, California, as of May 2024. The left figure assumes pure rate refinancing (non-cash out), while the right figure assumes a cash-out scenario. Both figures illustrate the tradeoff between upfront costs and interest rates.

Table B.7: Mean Values by Income Quintile

Income Quintile	Loan ID (count)	Credit Score	LTV (%)	PTI (%)	Initial Loan Balance ($)
1	90,331 <24.1>	744.3 (246.7)	69.1 (20.4)	39.5 (8.7)	113,079.9 (48,140.0)
2	78,577 <21.0>	746.9 (201.9)	72.8 (17.5)	36.9 (9.2)	180,094.7 (65,313.4)
3	73,251 <19.5>	750.1 (181.9)	73.4 (16.6)	35.1 (9.6)	236,640.2 (88,343.2)
4	68,386 <18.2>	756.5 (220.8)	73.1 (16.1)	32.7 (9.7)	299,712.5 (113,507.7)
5	64,332 <17.2>	765.6 (235.0)	69.9 (16.4)	25.5 (9.5)	368,110.3 (144,937.8)
Total	374,877 <100.0>	751.8 (219.6)	71.5 (17.8)	34.5 (10.4)	228,178.7 (130,095.4)

Income Quintile	Maturity (Month)	Income ($)	Monthly Payment ($)	House Value ($)
1	328.8 (66.1)	18,251.7 (5,230.7)	595.0 (205.7)	180,890.9 (106,661.4)
2	325.2 (68.7)	30,718.6 (4,457.3)	943.4 (263.5)	264,049.8 (132,821.6)
3	323.6 (70.0)	42,484.3 (5,611.8)	1,239.8 (366.4)	341,449.0 (172,616.7)
4	319.7 (72.8)	58,208.9 (8,015.1)	1,578.9 (488.9)	431,965.9 (214,241.4)
5	301.9 (82.9)	103,865.1 (58,277.9)	2,026.4 (715.0)	562,296.5 (305,448.9)
Total	320.8 (72.3)	47,369.5 (38,036.0)	1,214.4 (654.4)	339,725.3 (232,158.2)

Notes: The table reports means and standard deviations (in parentheses) by income quintile. The Loan ID column represents the unique number of loans, and the values within the angle brackets indicate the proportion of loans within that quintile relative to the total number of loans. Unconditional income quintiles are divided into 20% segments each, but the matched panel is based on matched refinanced loans. Therefore, the proportion of the matched panel can vary according to the number of refinanced loans in each income quintile.

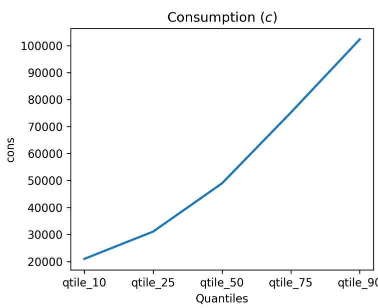
B.2 Statistic Moments of the Baseline Model

Table B.8: Interest Rate Reductions by Refinancing

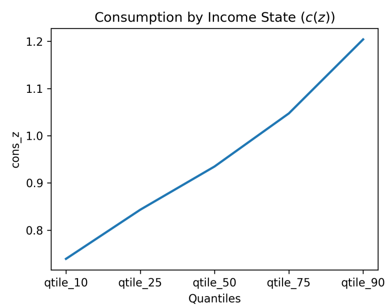
Dependent	Interest Rate Reduction by Refi (%p)				
Equation	(1)	(2)	(3)	(4)	(5)
IncomeQuintile 2	-0.186[a]	0.0108[a]	0.0166[a]	0.000147	0.000926
	(0.00317)	(0.00273)	(0.00293)	(0.00298)	(0.00296)
IncomeQuintile 3	-0.294[a]	0.0134[a]	0.0216[a]	-0.00529[c]	-0.00741[b]
	(0.00320)	(0.00283)	(0.00307)	(0.00319)	(0.00317)
IncomeQuintile 4	-0.370[a]	0.0141[a]	0.0249[a]	-0.0112[a]	-0.0176[a]
	(0.00328)	(0.00297)	(0.00324)	(0.00345)	(0.00343)
IncomeQuintile 5	-0.446[a]	0.0296[a]	0.0440[a]	-0.0133[a]	-0.0198[a]
	(0.00342)	(0.00319)	(0.00350)	(0.00402)	(0.00400)
Constant	2.283[a]	0.963[a]	0.949[a]	1.059[a]	1.045[a]
	(0.00229)	(0.00701)	(0.00709)	(0.00846)	(0.00843)
Control variables					
LTV, credit score, loan age, ZIP code		x	x	x	x
Credit tightness, HARP			x	x	x
Loan balance to Income				x	x
Est. equity					x
Observations	482,078	481,683	481,683	481,683	481,683
R-squared	0.043	0.351	0.352	0.354	0.361

	Predicted Interest Rate Reduction by Refi (%p)				
Equation	(1)	(2)	(3)	(4)	(5)
IncomeQuintile 1	2.2828	2.0202	2.0147	2.0405	2.0434
	(0.0023)	(0.0021)	(0.0021)	(0.0022)	(0.0022)
IncomeQuintile 2	2.0965	2.0310	2.0319	2.0415	2.0444
	(0.0022)	(0.0019)	(0.0019)	(0.0019)	(0.0019)
IncomeQuintile 3	1.9888	2.0337	2.0353	2.0345	2.0349
	(0.0022)	(0.0019)	(0.0019)	(0.0019)	(0.0019)
IncomeQuintile 4	1.9128	2.0344	2.0370	2.0268	2.0234
	(0.0024)	(0.0020)	(0.0020)	(0.0020)	(0.0020)
IncomeQuintile 5	1.8372	2.0498	2.0524	2.0204	2.0163
	(0.0025)	(0.0022)	(0.0022)	(0.0025)	(0.0025)

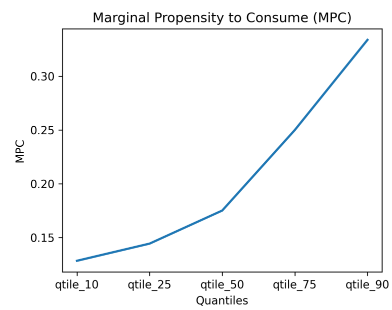
Notes: This table presents estimated interest rate savings through refinancing using Equation 5. The upper section of the table displays the regression results, while the lower section shows predicted values of the dependent variable based on the regression results. Superscripts a, b, c denote significance at 1%, 5%, 10% levels respectively. S.E. in parenthesis.



(a) Consumption

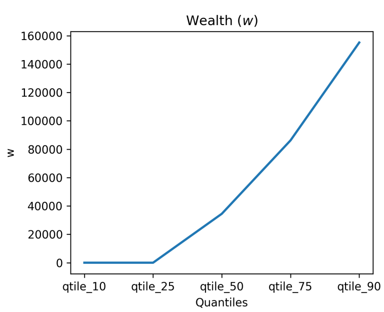


(b) Consumption to Income

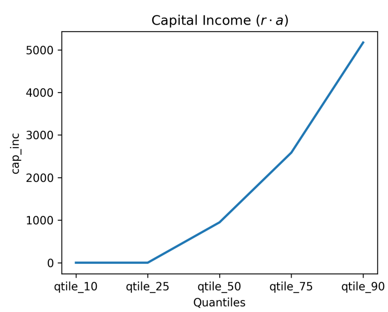


(c) MPC

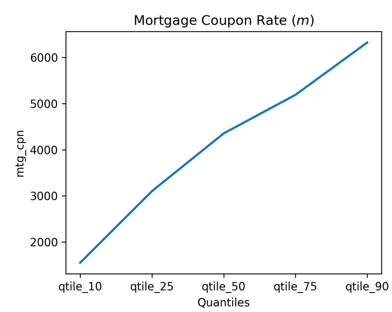
Figure B.4: Comparison across three cases



(a) Wealth



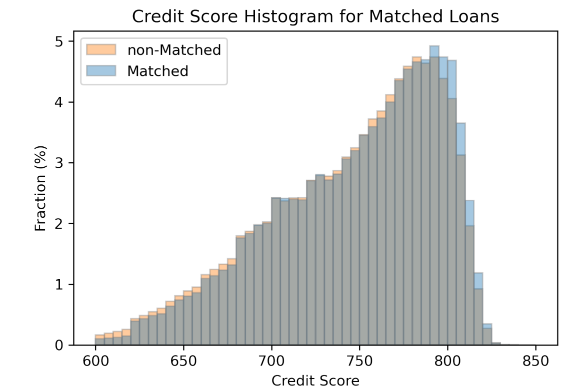
(b) Capital Income



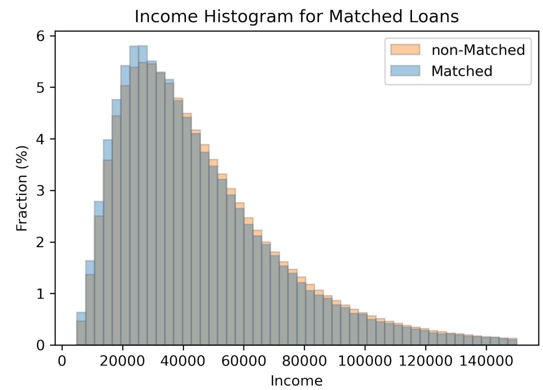
(c) Mortgage Coupon Payment

Figure B.5: Cross-Sectional Moments of the Baseline Model

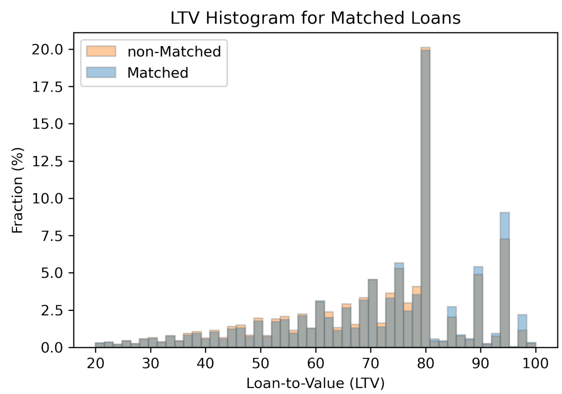
B.3 Data Validity Checks



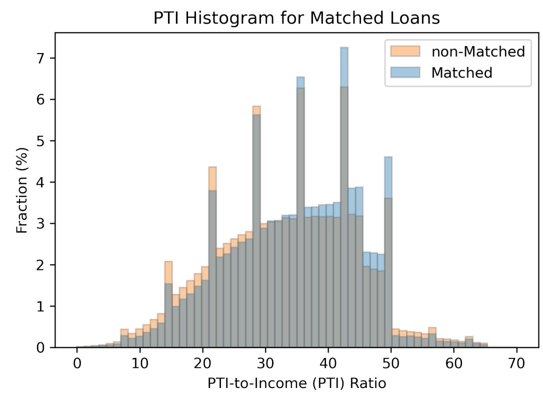
(a) Credit Score



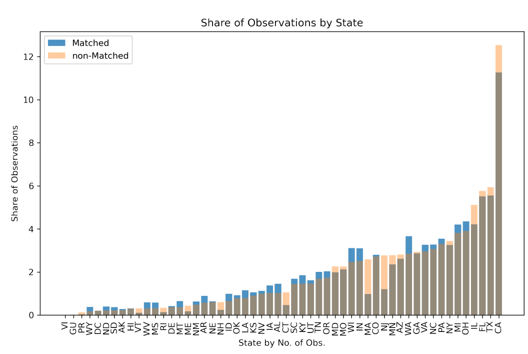
(b) Income



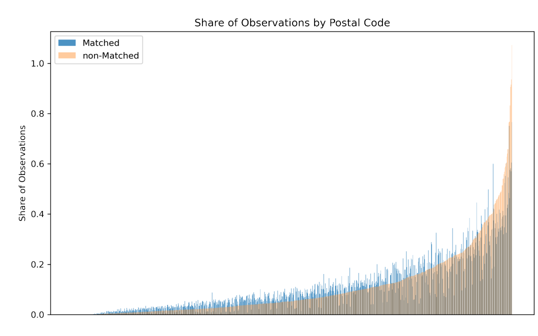
(c) Loan-to-Value (LTV)



(d) Payment-to-Income (PTI)



(e) Share of Observations by State



(f) Share of Observations by ZIP code

Figure B.6: Matched and Non-Matched Data Comparison by Distribution of Key Variables

Table B.9: Data Validity Checks on the Matched Panel

	Loan ID (count)	Credit Score	Loan Balance	Loan Term	Income	LTV	PTI
Matched Panel (mean)	374,877	751.8	228,179	320.8	47,369	71.5	34.5
All (mean)	41,150,001	758.4	213,022	309.4	49,514	70.1	33.3
Difference	*0.9% of total*	-0.9%	7.1%	3.7%	-4.3%	2.0%	3.5%
Matched Panel (median)	374,877	757.0	202,000	360.0	38,829	76.0	35.0
All (median)	41,150,001	754.0	184,000	360.0	40,474	75.0	34.0
Difference	*0.9% of total*	0.4%	9.8%	0.0%	-4.1%	0.0%	2.9%

C Appendix: Robustness Checks

C.1 Credit Score Bins

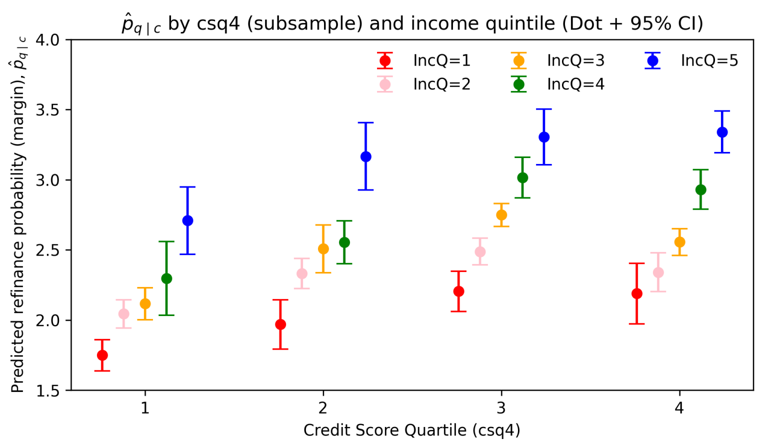


Figure C.7: Refinancing Propensity by Income Quintile and Leverage Ratio

Notes: The sample is split into quartiles of the borrower credit score. Within each credit-score quartile, we re-estimate the baseline linear probability model using the same set of controls and fixed effects as in the main specification. The figure reports the estimated refinancing probabilities for each income quintile within a given credit-score bin. Dots represent point estimates and bars denote 95

Table C.10: Income Gradient in Refinancing Within PTI Bins (ITM Sample)

PTI bin	< 36	$[36, 43)$	$[43, 45)$	$[45, 50)$	$[50, 60)$	≥ 60
Income Q2	0.388**	0.166	0.160	0.451**	0.466**	1.168**
	(0.109)	(0.105)	(0.182)	(0.131)	(0.143)	(0.308)
Income Q3	0.579**	0.370*	0.240	0.794**	1.028**	1.536*
	(0.122)	(0.164)	(0.249)	(0.187)	(0.229)	(0.708)
Income Q4	0.970**	0.378	0.095	0.930**	1.398**	1.396*
	(0.119)	(0.253)	(0.323)	(0.279)	(0.277)	(0.678)
Income Q5	1.409**	0.973**	1.549*	1.593**	1.214*	7.174**
	(0.153)	(0.284)	(0.640)	(0.360)	(0.557)	(1.521)

Notes: The sample is partitioned by payment-to-income (PTI) bins corresponding to common underwriting thresholds: PTI < 36, $[36, 43)$, $[43, 45)$, $[45, 50)$, $[50, 60)$, and ≥ 60. Within each PTI bin, we re-estimate the baseline linear probability model using the same set of controls and fixed effects as in the baseline specification. Reported coefficients correspond to income-quintile indicators relative to the lowest income quintile (Q1). Standard errors are reported in parentheses and significance levels are indicated by $* \, p < 0.10$, $** \, p < 0.05$, and $* * * \, p < 0.01$.

C.2 Changes in Income Quintile

Table C.11: Income Quintile Transition Matrix

Data	Income Quintile	To 1	To 2	To 3	To 4	To 5	Total (%)
	From 1	68.9	14.8	6.8	4.0	5.6	100.0
	2	26.2	35.7	22.6	9.6	6.0	100.0
PSID[1]	3	12.6	17.8	33.2	24.9	11.5	100.0
	4	4.7	7.6	17.2	38.6	31.9	100.0
	5	2.4	1.6	3.2	10.0	82.7	100.0
	From 1	43.6	24.6	15.0	10.0	6.8	100.0
	2	20.9	30.4	22.7	15.4	10.7	100.0
Loan Panel[2]	3	9.3	22.4	28.9	23.1	16.3	100.0
	4	4.9	11.4	22.8	33.4	27.5	100.0
	5	2.9	6.1	13.0	25.6	52.4	100.0

Notes: (1) Panel Study of Income Dynamics. Changes of mortgagor's income over two years, 2005–2021. (2) Changes from origination to refinancing.

Table C.12: Using New Loan's Income Quintile (Group Coeff. Only)

Dependent Equation	Refinancing Indicator ×100 (1)	(2)	(3)	(4)	(5)
Income quintile 2	0.286***	0.340***	0.336***	0.336***	0.286***
	(0.048)	(0.068)	(0.072)	(0.072)	(0.075)
Income quintile 3	0.573***	0.723***	0.731***	0.731***	0.573***
	(0.047)	(0.066)	(0.065)	(0.065)	(0.074)
Income quintile 4	0.857***	1.090***	1.104***	1.104***	0.860***
	(0.049)	(0.057)	(0.057)	(0.057)	(0.089)
Income quintile 5	1.165***	1.504***	1.522***	1.522***	1.211***
	(0.072)	(0.086)	(0.088)	(0.088)	(0.138)

Notes: Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
Controls by column: (1) none; (2) LTV bins, credit-score bins, loan-age bins, rate incentive; (3) (2) + credit tightness (denial_pctp, SLOOS×IncomeQ) + HARP×IncomeQ; (4) (3) + loan balance (UPB); (5) (4) + estimated LTV (appraisal-based LTV).

Table C.13: Restricting Samples Without Income Quintile Changes (Group Coeff. Only)

Dependent Equation	Refinancing Indicator ×100 (1)	(2)	(3)	(4)	(5)
Income quintile 2	0.269**	0.316**	0.303**	0.271**	0.276**
	(0.067)	(0.070)	(0.070)	(0.072)	(0.073)
Income quintile 3	0.581**	0.647**	0.623**	0.442**	0.450**
	(0.092)	(0.096)	(0.096)	(0.104)	(0.104)
Income quintile 4	0.942**	1.014**	0.987**	0.606**	0.611**
	(0.076)	(0.085)	(0.082)	(0.097)	(0.096)
Income quintile 5	1.527**	1.573**	1.538**	1.027**	1.023**
	(0.103)	(0.115)	(0.107)	(0.122)	(0.121)

Notes: Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
Controls by column: (1) none; (2) LTV bins, credit-score bins, loan-age bins, rate incentive; (3) (2) + credit tightness (denial_pctp, SLOOS×IncomeQ) + HARP×IncomeQ; (4) (3) + loan balance (UPB); (5) (4) + estimated LTV (appraisal-based LTV).

Table C.14: 5-year Rolling Window (Group Coeff. only)

Rolling window end year	2005	2007	2009	2011	2013	2015	2017	2019	2021
Income quintile 2	0.715***	0.373***	0.449***	0.434***	0.435***	0.396***	0.357*	0.248	-0.155
	(0.127)	(0.082)	(0.052)	(0.051)	(0.063)	(0.091)	(0.206)	(0.186)	(0.268)
Income quintile 3	0.853***	0.585***	0.762***	0.848***	0.793***	0.483***	0.384**	0.184	-0.303
	(0.150)	(0.072)	(0.067)	(0.061)	(0.081)	(0.108)	(0.180)	(0.279)	(0.319)
Income quintile 4	1.071***	0.729***	0.996***	1.104***	1.074***	0.591***	0.632***	0.115	-0.381
	(0.203)	(0.121)	(0.124)	(0.099)	(0.125)	(0.117)	(0.214)	(0.330)	(0.385)
Income quintile 5	1.674***	1.008***	1.307***	1.635***	1.594***	0.934***	0.888***	0.290	-0.304
	(0.214)	(0.108)	(0.139)	(0.117)	(0.153)	(0.177)	(0.286)	(0.400)	(0.487)

C.3 Survey of Consumer Finances (SCF)

Finally, I utilized the Survey of Consumer Finances (SCF) to examine whether the pattern of refinancing inequality could be observed. While the SCF does not provide panel data and is conducted as a triennial survey, it allows us to observe the year of the most recent refinancing for currently held mortgages. This enables us to derive annual refinancing rates.

Figure C.9 Panel (a) shows the refinancing rates by income quintile, calculated using pooled data from seven surveys conducted between 2004 and 2022. When compared to the Freddie Mac data, the SCF data similarly reveals that lower-income households exhibit a lower propensity to refinance. However, the SCF data overall shows higher refinancing rates, likely due to differences in data coverage, as will be discussed later.

Conversely, Panel (b) displays refinancing rates divided into quintiles based on the leverage (loan balance to income) ratio as per Chen et al. (2020). Interestingly, this presents a contrasting pattern: the SCF data indicates higher refinancing activity as the leverage ratio increases, whereas the Freddie Mac data shows a decrease in refinancing propensity.

The divergence in these results can be attributed to two main factors. First, the nature of the datasets differs significantly. The SCF is a survey-based triennial dataset that uses current household income, whereas the Freddie Mac data is an administrative panel dataset based on the borrower's income at the time of loan origination. Consequently, differences in the calculation of the loan-to-income ratio, due to variations between household and individual income and current versus past income, complicate direct comparisons.

More importantly, the scope of households covered by each dataset appears to contribute to these contrasting results. As shown in Figure C.10, the types of mortgages in the U.S. are broadly categorized into conventional and non-conventional mortgages. Conventional mortgages are further divided into conforming and jumbo (non-conforming) loans. Conforming loans meet the guidelines set by the Federal Housing Finance Agency (FHFA), including loan limits and credit requirements, whereas jumbo loans exceed these limits.

Table C.15: Regression by Loan Age Restrictions

Dependent	Refinancing Indicator x 100			
Loan Age ≤	1 yr	2 yrs	3 yrs	5 yrs
IncomeQuintile 2	0.561[a]	1.220[a]	1.229[a]	0.962[a]
	(0.245)	(0.146)	(0.0951)	(0.0732)
IncomeQuintile 3	1.235[a]	1.963[a]	2.041[a]	1.741[a]
	(0.250)	(0.153)	(0.102)	(0.0783)
IncomeQuintile 4	2.192[a]	2.875[a]	2.752[a]	2.403[a]
	(0.259)	(0.161)	(0.111)	(0.0854)
IncomeQuintile 5	2.912[a]	3.642[a]	3.654[a]	3.283[a]
	(0.292)	(0.180)	(0.127)	(0.0981)
Constant	0.276	-0.373	-0.244	-0.0230
	(0.589)	(0.354)	(0.263)	(0.200)
Observations	97,515	374,996	670,264	945,099
R-squared	0.022	0.015	0.014	0.012

Notes: a, b, c denote significance at 1%, 5%, 10% levels respectively. S.E. in parenthesis. The results base on the benchmark regression (5); control variables are LTV, credit score, rate incentive, credit tightness, HARP, loan balance to income, estimated LTV.

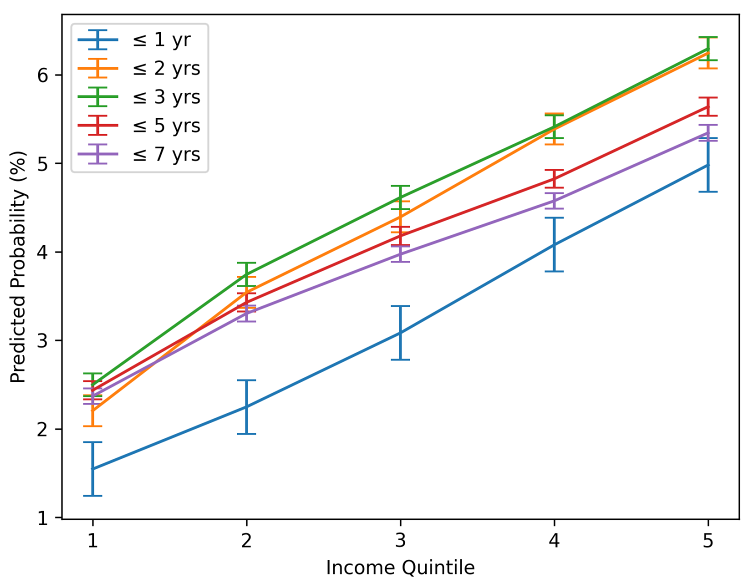
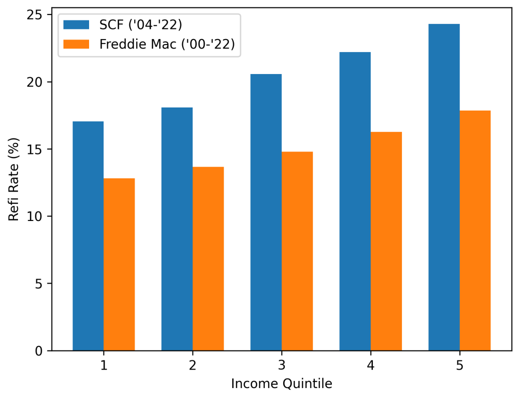
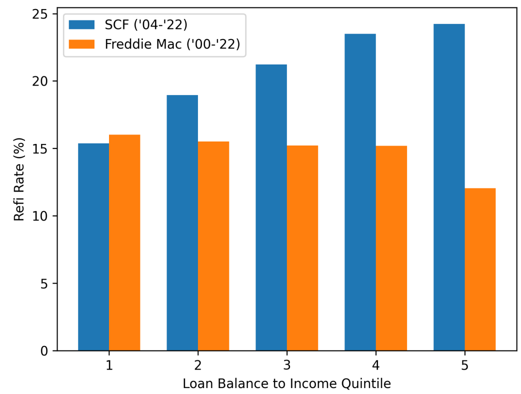


Figure C.8: Predicted Refinancing Probability by Loan Age

Notes: This plot shows the predicted monthly refinancing probability by loan age, using Equation 2. The results are based on the benchmark regression (5); control variables are LTV, credit score, rate incentive, credit tightness, HARP, loan balance to income, and estimated LTV. The 95% confidence intervals are represented by error bars.



(a) Refinancing Rate by Income



(b) Refinancing Rate by Leverage Ratio

Figure C.9: Refinancing Propensity by Income Quintile and Leverage Ratio

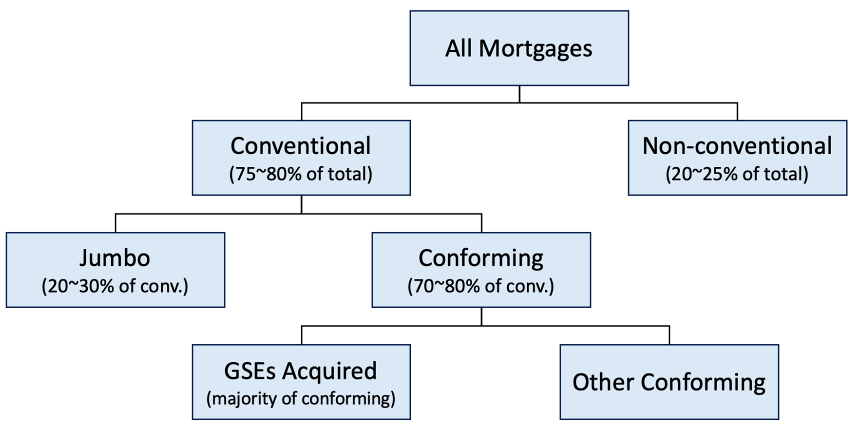


Figure C.10: Mortgage Types in the U.S.

Notes: Market shares of each mortgage type vary over time, and the numbers provided are approximate, referencing the National Mortgage Database (NMDB), web searches.

Conforming loans are mostly acquired in the secondary market by government-sponsored enterprises (GSEs) like Fannie Mae and Freddie Mac. The Freddie Mac data that I used primarily represents about 70% of conventional loans, accounting for approximately 60% of all loans.

However, the loans not covered by the Freddie Mac data, particularly the jumbo loans, which constitute about 30% of conventional loans, seem to be influencing the results. Jumbo loans, exceeding conforming loan limits, are predominantly held by high-income, wealthy individuals. Since the SCF samples all households, including jumbo loan holders, it exhibits significantly higher income levels, even considering household versus borrower income as shown in Table C.16.

Table C.16: Freddie Mac (Borrower) and SCF (Household) Income by Leverage Ratio

Loan-to-income Quintile	Median Income ($)		Mean Income ($)	
	Freddie Mac	SCF	Freddie Mac	SCF
1	53,143	356,204	68,713	1,790,221
2	41,095	161,175	47,831	366,883
3	35,438	119,000	40,494	189,615
4	31,529	95,000	35,493	142,475
5	28,984	58,396	31,971	83,983

Notes: It is important to note that the SCF uses current household income, while Freddie Mac uses the borrower's individual income at the time of loan origination. The statistics are based on the SCF 2004-2022 triannual public data by pooling it together, and Freddie Mac used a sample based on loans originated from 2000 to 2022.

Thus, assuming comparability despite the inherent differences in data characteristics, the sample including all households shows increased refinancing with higher leverage. In contrast, narrowing the analysis to middle and lower-income households reveals an opposite and intriguing pattern. Highly leveraged households might have a high demand for cash-out refinancing due to idiosyncratic shocks, but those with higher absolute incomes might experience fewer frictions related to low income, such as liquidity constraints for closing costs or financial illiteracy, which affect refinancing decisions.

Regarding policy relevance, which dataset is more relevant? If the limitations of survey data is disregard, the SCF offers a more comprehensive coverage. However, if monetary policy focuses on relatively liquidity-constrained, high-MPC middle and lower-income households, the subset I used would be more significant for policymakers. Specifically, frictions unique to lower-income households can disrupt the transmission channels of monetary policy, making the refinancing behavior of the wealthy less pertinent. Comparing refinancing propensity using consistent income standards across different data scopes could be an interesting area for further research.

C.4 Estimating Unrealized Savings by Constructing Counterfactual Rate Gaps

As an alternative approach to quantifying unrealized savings, I estimate income-group-specific rate gaps directly from the population of in-the-money loan-months, rather than relying on the matched-loan sample. This method constructs a counterfactual refinancing rate for each borrower and translates it into an implied monthly payment reduction.

Step 1: Estimating the predicted rate gap by income group. I regress the rate incentive $RateGap_{it} \equiv c_{it} - m_{it}$ on income quintile indicators and the full set of controls, restricting the sample to in-the-money observations ($gap > 0$):

$$RateGap_{it} = \alpha + \sum_{q=2}^{5} \hat{\beta}_q \cdot \mathbf{1}\{IncomeQ_i = q\} + \Gamma' \cdot X_{it} + \nu_{z(i)} + \lambda_{v(i)} + \epsilon_{it} \tag{22}$$

where c_{it} is borrower i's contracted coupon rate and m_{it} is the prevailing obtainable rate. The coefficients $\hat{\beta}_q$ capture the income-quintile gradient in the rate incentive after controlling for borrower and loan characteristics, ZIP fixed effects, and vintage fixed effects.

Step 2: Constructing the counterfactual refinancing rate. Using the estimated income-group predicted rate gap $\widehat{RateGap}_{q(i)}$, I construct a counterfactual new coupon rate for each in-the-money loan-month:

$$\hat{m}_{it}^{new} = m_{it}^{*} - \widehat{RateGap}_{q(i)} \tag{23}$$

This counterfactual rate represents what a borrower in income quintile q would pay if they refinanced and captured the income-group-specific rate benefit. I then translate this into an implied monthly payment reduction using loan-specific balance B_{it}, current coupon c_{it}, and remaining term N_{it}:

$$\widehat{\Delta Pay}_{it} = PMT(B_{it}, m^*_{it}, N_{it}) - PMT\left(B_{it}, \hat{m}^{new}_{it}, N_{it}\right) \tag{24}$$

Step 3: Aggregating unrealized savings by income group. I aggregate $\widehat{\Delta Pay}_{it}$ across in-the-money loan-months within each income quintile and normalize by average income to obtain the payment reduction as a share of income.

The results are reported in Table C.17. Consistent with the matched-loan estimates, the implied monthly payment reduction is largest relative to income for the lowest-income borrowers, amounting to approximately 7.6% of monthly income for the bottom quintile compared with 4.5% for the top quintile. The similarity of these estimates to those from the matched-loan approach corroborates the robustness of the main finding: foregone refinancing savings are disproportionately large relative to income for lower-income households, precisely those with the highest marginal propensity to consume.

Income Quintile	Avg. Income ($)	Avg. Balance ($)	Est. Rate Gap (pp)	Mthly. Pymt. Reduction ($)	Reduction / Income (%)
1	16,579	89,989	1.01	100.14	7.61
2	28,121	145,956	1.00	158.26	6.58
3	39,391	195,149	0.99	207.84	6.12
4	54,772	253,904	0.97	268.55	5.56
5	98,047	312,712	0.96	379.16	4.49

Payment savings assume refinancing resets the loan to a 30-year FRM term. Rate gaps are estimated using Eq. (5) (full controls). Freddie Mac statistics suggest that about 70–75% of refinances keep the term length unchanged. Sample is restricted to in-the-money observations ($gap > 0$).

Table C.17: Unrealized savings under the counterfactual

C.5 Different Specifications of Refinancing Response

Apart from Equation 19, there may be other specifications to measure regional refinancing responses. Therefore, in this section, I verify robustness by using alternative methods.

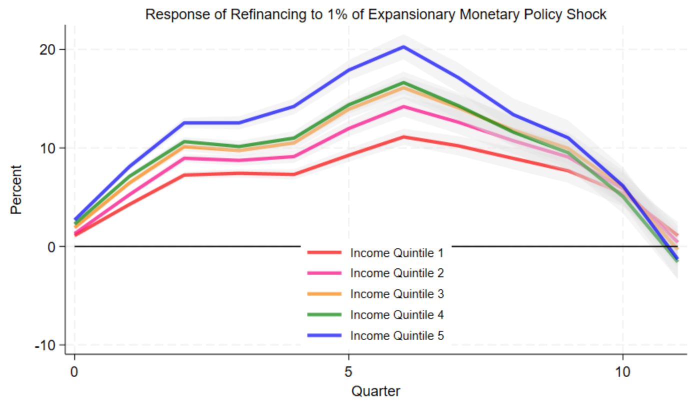
$$\rho_{t+h}^{r,q} = \frac{x_t^{r,q} + \ldots + x_{t+h}^{r,q}}{y_{t-1}^{r,q}} \tag{25}$$

$$\rho_{t+h}^{r,q} = \frac{x_t^{r,q}}{y_t^{r,q}} + \ldots + \frac{x_{t+h}^{r,q}}{y_{t+h}^{r,q}} \tag{26}$$

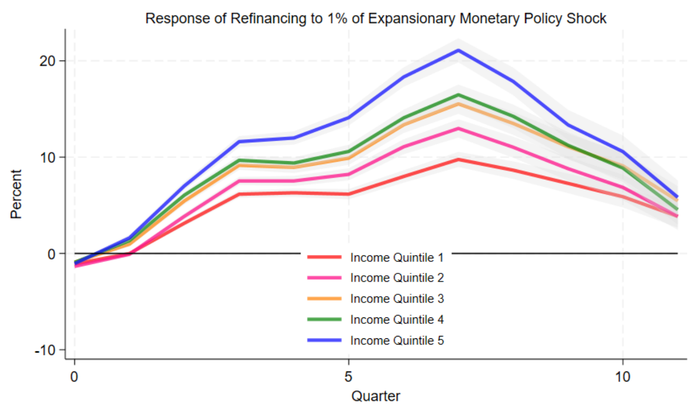
Again, $x_t^{r,q}$ is the number of refinancing loans by income quintile q in region r in quarter t, $y_t^{r,q}$ is the number of all outstanding loans, and h is the analysis time horizon.

Equation 25 fixes the denominator to the period before the monetary shock arrives. This approach reduces the distortion caused by underestimating the refinancing share due to an increase in the total number of outstanding loans prompted by home purchases in that region. On the other hand, if the horizon extends and there are loans that refinance multiple times, it may overestimate the response magnitude. Equation 26 is the cumulative sum of the refinancing ratios for each quarter. If the total number of outstanding mortgages does not change significantly, it is expected to show a ratio similar to Equation 25.

Both specifications consistently demonstrate refinancing inequality. Although the timing of the peak differs by only one quarter, both indicate that high-income households refinance significantly more than low-income households.



(a) Equation 25



(b) Equation 26

Figure C.11: Refinancing Response to 1% Interest Rate Cut by Income Quintile

Notes: Panel (a) shows the cumulative refinancing response by income quintile to 1% interest rate falls using Equation 26. The dependent variable is the cumulative proportion of loans refinanced in each region and income group as described in Equation 5. Regions are 908 three-digit ZIP code, and quarterly data from 2004 to2022was used

D Appendix: Identification of In-the-money Loans

As described in the text, an in-the-money loan refers to a situation where the difference between the old and new interest rates after refinancing exceeds a certain threshold. This threshold represents the fixed costs associated with refinancing and option value.

$$\text{In-the-money } \textit{if } \underbrace{(\text{Interest Rate}_i^{old} - \text{Potential Refi Rate}_{i,t}^{new}) - \text{Threshold}_{i,t}}_{\equiv \text{ Rate Incentive}} > 0$$

(i and t denote loan and time respectively)

Agarwal et al. (2013) has proposed the following tractable closed-form solution for the threshold value.

$$\begin{aligned} Threshold &= \frac{1}{\psi}\left[\phi + W\left(-\exp\left(-\phi\right)\right)\right] \\ \psi &= \sqrt{\frac{2(\rho+\lambda)}{\sigma}} \\ \phi &= 1 + \psi(\rho+\lambda)\frac{\kappa(M)/M}{(1-\tau)} \end{aligned}$$

Here, $W(\cdot)$ is the Lambert W-function, ρ is the real discount rate, λ is the expected real rate of exogenous mortgage repayment, σ is the standard deviation of the mortgage rate, $\kappa(M)/M$ is the ratio of the tax-adjusted refinancing cost and the remaining mortgage value, and τ is the marginal tax rate. The values of the parameters suggested by Agarwal et al., 2013 are $\sigma = 0.0109$, $\tau = 0.28$, $\rho = 0.05$.

$\kappa(M)$ is the cost entailed in refinancing, which is defined by:

$$\kappa(M) = F + fM\left[1 - \frac{\tau}{\theta+\rho+\pi}\left\{\frac{1-\exp\left(-(\theta+\rho+\pi)N\right)}{N}\frac{\rho+\pi}{\theta+\rho+\pi} + \theta\right\}\right]$$

F is the fixed cost of refinance, f is the points divided by 100, τ is the marginal tax rate, θ is the expected arrival rate of a full deductibility event, such as moving and refinancing. Using the parameter suggested by the paper $F = 2000$, $f = 0.01$, $\theta = \mu + 0.1 = 0.2$, the cost can be approximated as follows.

$$\kappa(M) = 2000 + 0.007905M$$

Lastly, Lambda is defined as the expected rate of decline in the real principal of the mortgage for reasons other than rate-reducing refinancing, and is defined as:

$$\lambda = \mu + \frac{i^{old}}{\exp(i^{old}\Gamma) - 1} + \pi,$$

Γ is the remaining life of the existing mortgage in years, i^{old} is the current (old) mortgage interest rate for existing loan, μ is the hazard of relocation, and π is the inflation rate. The parameter values used are $\mu = 0.1$ and $\pi = 0.03$, following the paper.

References

Agarwal, Sumit, John C. Driscoll, and David I. Laibson, "Optimal Mortgage Refinancing: A Closed-Form Solution: MONEY, CREDIT AND BANKING," *Journal of Money, Credit and Banking*, June 2013, *45* (4), 591–622.

_, Richard J. Rosen, and Vincent Yao, "Why Do Borrowers Make Mortgage Refinancing Mistakes?," *Management Science*, 2016, *62* (12), 3494–3509.

_, Souphala Chomsisengphet, Hua Kiefer, Leonard C. Kiefer, and Paolina C. Medina, "Refinancing Inequality During the COVID-19 Pandemic," *Journal of Financial and Quantitative Analysis*, August 2024, *59* (5), 2133–2163.

Amberg, Niklas, Thomas Jansson, Mathias Klein, and Anna Rogantini Picco, "Five Facts about the Distributional Income Effects of Monetary Policy Shocks," *American Economic Review: Insights*, September 2022, *4* (3), 289–304.

Andersen, Asger Lau, Niels Johannesen, Mia Jørgensen, and José-Luis Peydró, "Monetary Policy and Inequality," *The Journal of Finance*, 2023, *78* (5), 2945–2989. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/jofi.13262.

Andersen, Steffen, John Y. Campbell, Kasper Meisner Nielsen, and Tarun Ramadorai, "Sources of Inaction in Household Finance: Evidence from the Danish Mortgage Market," *American Economic Review*, October 2020, *110* (10), 3184–3230.

Auclert, Adrien, "Monetary Policy and the Redistribution Channel," *American Economic Review*, June 2019, *109* (6), 2333–2367.

Beraja, Martin, Andreas Fuster, Erik Hurst, and Joseph Vavra, "Regional Heterogeneity and the Refinancing Channel of Monetary Policy*," *The Quarterly Journal of Economics*, February 2019, *134* (1), 109–183.

Berger, David, Konstantin Milbradt, Fabrice Tourre, and Joseph Vavra, "Mortgage Prepayment and Path-Dependent Effects of Monetary Policy," *American Economic Review*, September 2021, *111* (9), 2829–2878.

Berger, David W., Konstantin Milbradt, Fabrice Tourre, and Joseph Vavra, "Refinancing Frictions, Mortgage Pricing and Redistribution," January 2024.

Byrne, Shane, Kenneth Devine, Michael King, Yvonne McCarthy, and Christopher Palmer, "The Last Mile of Monetary Policy: Inattention, Reminders, and the Refinancing Channel," March 2023.

Chen, Hui, Michael Michaux, and Nikolai Roussanov, "Houses as ATMs: Mortgage Refinancing and Macroeconomic Uncertainty," *The Journal of Finance*, 2020, *75* (1), 323–375. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/jofi.12842.

Coibion, Olivier, Yuriy Gorodnichenko, Lorenz Kueng, and John Silvia, "Innocent Bystanders? Monetary policy and inequality," *Journal of Monetary Economics*, June 2017, *88*, 70–89.

Defusco, Anthony A. and John Mondragon, "No Job, No Money, No Refi: Frictions to Refinancing in a Recession," *The Journal of Finance*, 2020, *75* (5), 2327–2376. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/jofi.12952.

Defusco, Anthony A, Stephanie Johnson, and John Mondragon, "Regulating Household Leverage," *The Review of Economic Studies*, August 2019, p. rdz040.

Eichenbaum, Martin, Sergio Rebelo, and Arlene Wong, "State-Dependent Effects of Monetary Policy: The Refinancing Channel," *American Economic Review*, March 2022, *112* (3), 721–761.

Fisher, Jack, Alessandro Gavazza, Lu Liu, Tarun Ramadorai, and Jagdish Tripathy, "Refinancing cross-subsidies in the mortgage market," *Journal of Financial Economics*, August 2024, *158*, 103876.

Floden, Martin and Jesper Lindé, "Idiosyncratic Risk in the United States and Sweden: Is There a Role for Government Insurance?," *Review of Economic Dynamics*, 2001, *4* (2), 406–437.

Gerardi, Kristopher, Paul S. Willen, and David Hao Zhang, "Mortgage prepayment, race, and monetary policy," *Journal of Financial Economics*, March 2023, *147* (3), 498–524.

Jørring, Adam and Ryan Westphal, "How Important is Household Heterogeneity for Monetary Policy? Evidence from Mortgage Refinancing," February 2025.

Kaplan, Greg, Benjamin Moll, and Giovanni L. Violante, "Monetary Policy According to HANK," *American Economic Review*, March 2018, *108* (3), 697–743.

Keys, Benjamin J., Devin G. Pope, and Jaren C. Pope, "Failure to refinance," *Journal of Financial Economics*, December 2016, *122* (3), 482–499.

Maggio, Marco Di, Amir Kermani, and Christopher J Palmer, "How Quantitative Easing Works: Evidence on the Refinancing Channel," *The Review of Economic Studies*, May 2020, *87* (3), 1498–1528.

_ , _ , Benjamin J. Keys, Tomasz Piskorski, Rodney Ramcharan, Amit Seru, and Vincent Yao, "Interest Rate Pass-Through: Mortgage Rates, Household Consumption, and Voluntary Deleveraging," *American Economic Review*, November 2017, *107* (11), 3550–3588.

Mian, Atif and Amir Sufi, "The Consequences of Mortgage Credit Expansion: Evidence from the U.S. Mortgage Default Crisis*," *The Quarterly Journal of Economics*, November 2009, *124* (4), 1449–1496.

Nakamura, Emi and Jón Steinsson, "High-Frequency Identification of Monetary Non-Neutrality: The Information Effect*," *The Quarterly Journal of Economics*, August 2018, *133* (3), 1283–1330.

Swanson, Eric T., "Measuring the effects of federal reserve forward guidance and asset purchases on financial markets," *Journal of Monetary Economics*, March 2021, *118*, 32–53.

Zhang, David Hao, "Closing Costs, Refinancing, and Inefficiencies in the Mortgage Market," January 2022.

Òscar Jordà, "Estimation and Inference of Impulse Responses by Local Projections," *American Economic Review*, March 2005, *95* (1), 161–182.